UNIVERSITY OF ECONOMICS AND LAW

FACULTY OF INFORMATION SYSTEM

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FINAL PROJECT REPORT

MACHINE LEARNING IN BUSINESS ANALYTICS

TOPIC:

Developing a Multidimensional Machine Learning Integrated Application for Small Businesses

Lecturer: **Ph.D. Trần Duy Thanh**

Student:

Huỳnh Huệ Trúc – K224111475

Thái Anh Thư - K224111468

Lê Huyền Trân - K224111472

Hồ Song Tín - K224111469

Nguyễn Tiến Đạt - K224060830

Nguyễn Quang Minh - K224060841

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Lastly, we express our sincere appreciation to the professors, doctors, and authors whose relevant research provided essential information and valuable data throughout our research process. Your contributions have been instrumental in shaping our work and ensuring its success.

Commitment

Our group, under the guidance of Assoc. Prof. Dr. Tran Duy Thanh, affirms that all content within this project has been developed by us, with the support and supervision of our advisor. Utilizing research results based on realistic business operations, we have created simulated data that aligns with the objectives and hypotheses of our study. We ensure the clarity, legality, and proper acknowledgment of all data, research results, relevant documents, and references. We take full responsibility for the accuracy and authenticity of our work.

As this is our first project in machine learning, we deeply understand that there are areas for improvement. Therefore, we welcome your feedback and suggestions to enhance the quality of our research. Your insights are invaluable for our future endeavors in this field.

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Introduction

In today's fast-paced global economy, retail businesses are confronted with formidable challenges in achieving growth and optimization. The landscape is shaped by volatile market conditions and the widespread adoption of e-commerce platforms, intensifying the struggle for customer retention. Faced with these realities, businesses are compelled to adapt rapidly, with many turning to advanced technologies like machine learning to fortify their strategic responses and bolster competitiveness.

According to Rackspace Technology's "The 2023 AI and Machine Learning Research Report," a staggering 72% of surveyed participants, totaling over 1,400 individuals, have integrated AI and machine learning into their business and IT strategies. Notably, 69% of respondents identified AI/ML as a high-priority technology, leveraging it to refine existing processes, forecast business performance and industry trends, and mitigate risks.

Sales forecasting through machine learning enables businesses to discern shopping trends and customer behavior from vast datasets, facilitating accurate predictions of future product demand (Chase, 2016). These insights serve as the bedrock for estimating sales revenue and formulating informed decisions regarding production, operations, and marketing strategies (Marshall et al., 2013; Shi et al., 2015).

Machine learning recommendation systems have become integral in helping users discover new items or services, leveraging user information and item attributes to deliver personalized suggestions (Adomavicius & Tuzhilin, 2005). These systems enhance decision-making, maximizing profits and minimizing risks (Chen et al., 2008; Bouneffouf et al., 2013). They are widely used in many information-based companies such as Google (Liu, Dolan, & Pedersen, 2010), Twitter (Ahmed et al., 2013), LinkedIn (Rodriguez, Posse, & Zhang, 2012), and Netflix (Steck, 2013) to improve user engagement and satisfaction.

However, the seamless integration of advanced technologies such as machine learning into retail operations is often impeded by a host of challenges. These include technological barriers, prohibitive implementation costs, a scarcity of skilled professionals, and resistance to change within organizational cultures (Heins, 2023). Overcoming these hurdles is paramount, as failure to do so can result in stagnant management practices, erratic business performance, and compromised competitive sustainability.

To bridge the gap between economic progress and cutting-edge technology, machine learning has been woven into the fabric of modern business operations and is poised to become an indispensable component of business intelligence. In our study, we present a multidimensional approach aimed at elevating sales and business operating within businesses through the application of machine learning. Our approach encompasses both ends of the spectrum: a recommendation system empowering customers to make tailored product choices and a sales forecasting system equipping managers to optimize stock levels and sales strategy based on predictive insights.

Our suite of five sales forecasting models empowers business administrators to effortlessly train, deploy, and evaluate machine learning models tailored to their datasets. Simultaneously, our recommendation system comprises three distinct types—collaborative, content-based, and hybrid—seamlessly delivering real-time product suggestions to both new and returning customers based on their individual preferences.

In conclusion, we firmly believe that the successful integration of machine learning into retail operations has the potential to revolutionize decision-making processes, elevate customer satisfaction, and confer a significant competitive advantage. With a steadfast commitment to overcoming technological barriers and harnessing the power of sophisticated data analytics, we aspire to offer our assistance, not only to small businesses individually but also to the broader retail industry. Together, we aim to achieve sustained growth and heightened operational efficiency.

Objectives

The objective of developing this machine learning-integrated application is to enhance the efficiency of small stores in managing their data and customer interactions. Our design aims to support various aspects of business operations. By forecasting sales, we help business admins with inventory management and optimizing pricing strategies to increase profitability. It also seeks to improve customer relationship management by personalizing marketing efforts and boosting customer loyalty. Furthermore, the application enhances data-driven decision-making for both sellers and buyers, enabling better resource allocation, demand planning, and effective selling. This targeted approach ensures that small shop owners can operate more efficiently, maintain a competitive edge in the market, and achieve high levels of customer satisfaction.

Customers: Customers can effortlessly place orders through a distinct GUI, enjoying a seamless, simple, and convenient shopping experience. Each time a user opens the app, a tailored recommendation system automatically displays relevant products to their interest, making it easy for customers to find and purchase items they love. Additionally, the inclusion of three different recommendation systems—collaborative, content-based, and hybrid—

ensures that both new and returning customers can discover products whether they have specific items in mind or not.

Employees: This app empowers the employee team to efficiently manage and track essential information across various entities. It allows for comprehensive management of customers, employees, and products, alongside the ability to monitor ingredients and store recipes. Additionally, employees can update and track the status of customer orders in real time. With these features, the app ensures streamlined operations and up-to-date information, enhancing productivity and coordination within the team.

Managers: The app harnesses the power of machine learning to equip managers with invaluable insights. By analyzing sales data through statistics and forecasting models, managers can discern patterns and trends, empowering them to refine sales and marketing strategies effectively. This data-driven approach enables managers to make informed decisions, ultimately driving improvements in sales performance and marketing effectiveness.

Subjects and scopes

Research subjects:

This research aims to develop and implement machine learning models to optimize workforce management, refine sales and marketing strategies, and enhance customer experience in beverage stores. By focusing on predictive scheduling, performance analytics, campaign effectiveness, sales forecasting, and personalized customer services, the study seeks to demonstrate how machine learning can drive significant improvements in operational efficiency, customer satisfaction, and overall business performance

Research scope:

This research explores the integration of machine learning to enhance operational efficiency, customer satisfaction, and overall business performance in beverage stores. It focuses on three critical areas: workforce management and optimization, sales and marketing analytics, and customer experience enhancement.

1. Workforce Management and Optimization

Efficient workforce management is vital for the smooth operation of beverage stores. This research will develop machine learning models that leverage predictive scheduling and performance analytics to optimize staff allocation. The models will analyze demand forecasts and historical performance data to create algorithms that predict peak hours and customer flow, ensuring adequate staffing levels while minimizing labor costs. Additionally, performance analytics will identify training needs and reward high-performing employees, leading to a more motivated and efficient workforce.

2. Sales and Marketing Analytics

Effective sales and marketing strategies are essential for driving revenue and customer engagement in beverage stores. This research will employ machine learning techniques to analyze and optimize marketing campaigns. Through A/B testing, campaign effectiveness analysis, and sales forecasting, the research will provide insights into the most effective strategies. This will enable managers to tailor their campaigns for maximum impact, enhancing the return on marketing investments and ensuring promotional efforts align with customer preferences and market trends.

3. Customer Experience Enhancement

Delivering a superior customer experience is crucial for retaining customers and fostering loyalty. This research will leverage machine learning for customer journey analytics, service personalization, and experience management. By analyzing customer interactions and feedback, the models will identify key touchpoints and pain points in the customer journey. This information will be used to personalize services, recommend products based on past purchases and preferences, and improve overall store interactions. Enhancing customer satisfaction through tailored experiences will boost customer loyalty and generate positive word-of-mouth, attracting new customers.

Time Scope: From 4/2022 to 4/2024.

By focusing on workforce management and optimization, sales and marketing analytics, and customer experience enhancement, this research seeks to harness the power of machine learning to drive significant improvements in operational efficiency and customer satisfaction in beverage stores.

Space Scope: The study will consider all orders that arose within the specified time scope.

Business challenges

Business problem

Resource allocation is a critical aspect of business management that directly impacts a company's ability to grow and compete in the market. Effective resource allocation involves distributing the company's resources—such as personnel, capital, and inventory—in a manner that maximizes productivity and profitability. Proper resource allocation ensures that resources are used where they are most needed, reducing waste and improving overall efficiency. Therefore, As businesses grow, having an accurate sales forecast allows them to scale operations smoothly without experiencing bottlenecks or resource shortages. According to a report by the Harvard Business Review, companies that implement effective sales forecasting methods can reduce inventory costs by up to 20% and increase sales by up to 15% due to better resource management and availability of products.

The primary goal of any business is to optimize inventory levels while enhancing customer service and satisfaction. Accurate sales forecasts enable businesses to anticipate demand, seize market opportunities, and respond promptly to customer needs (Shireen, 2023). Poor sales forecasting can adversely impact various aspects of inventory management, including costs, revenue, and operational efficiency. Furthermore, businesses that fail to accurately forecast sales may face significant challenges in maintaining the right inventory levels, which can lead to either excess stock or stockouts. Both scenarios are detrimental: excess stock ties up capital and increases holding costs, while stockouts result in lost sales and damage to customer trust. Effective inventory management, therefore, hinges on precise sales forecasting to balance these dynamics and ensure a smooth operation.

Moreover, understanding and capturing user psychology is fundamental to the effectiveness of recommendation systems. Accurately predicting user preferences requires a deep understanding of consumer behavior, which can be complex and influenced by various factors such as personal tastes, cultural trends, and social influences.(Zhang, Y., & Chen, X., 2020). Machine learning models need to be continuously updated and refined to adapt to changing user behaviors and preferences.

Decision-making, especially in inventory management, is crucial for the effective functioning of businesses. Stores must balance supply and demand, control costs, and ensure customer satisfaction by maintaining optimal stock levels. Advanced data analysis tools and software are used to predict demand and make timely restocking decisions. Avoiding overstocking and stockouts, thus improving efficiency and customer service. (Van Kampen et al., 2021), Moreover, the integration of technology, such as Machine Learning/AI, further streamlines inventory management processes, reducing errors and saving time (Hindawi, 2023). By making informed and strategic decisions, shop owners can optimize inventory levels, reduce costs, and boost customer satisfaction.

Market challenges

According to Coresight Research, as the retail industry embraces upcoming technology trends such as machine learning/AI and innovations in inventory management, it becomes imperative that our apps are not only well-performing to attract users and build their trust but also user-friendly for retail owners who may lack a strong technological background.

Data challenges

One of the foremost challenges in retail sales forecasting is ensuring the quality and quantity of data. Administrators must handle vast amounts of transactional data, which can often be incomplete or erroneous. Poor data quality can severely impact the accuracy of predictive models, leading to misguided forecasts that can result in overstock or understock situations. A well-behaved performance model requires an extremely huge amount of data, thus as the data accumulates, a more well-behaved performance model can be obtained.(Najafabadi et al. 2015).

Chapter 1. Data Simulation and Understanding

In this chapter, we will introduce and conduct a fundamental analysis of the data, as well as establish the foundation for data simulation. We will discuss the key principles and references that justify the use of simulated data in our sales forecasting model. This approach ensures that our analysis is robust, comprehensive, and based on real-world insights.

1.1. Data context

1.1.1. Data simulation facility

Given that the goal of this project is to develop a multi-dimensional user interface tailored for small businesses, we decided to construct our project database using business simulation tools. This approach ensures comprehensive coverage of all individuals and processes involved in the business. Consequently, our database will encompass information about key stakeholders and activities, including management (users who are Managers of the small business), purchasing (users who are Customers of the small business), and sales (users who are Employees of the small business).

Among the various research sources and references we reviewed, those chosen are expected to be the most important and popular facilities.

Table 1.1: Data simulation facility

No.	Rule	Reference
1	If the retail price of a product is high, th quantity of products sold will be lower.	Chen et al. (2011) and Li and Hi (2010)
2	If there are promotional programs in an month, the number of orders in that month ma be higher.	Yin and Huang (2014)
3	Ensure that orders are distributed unevenly meaning there are periods with more order than others.	The research team has observed an established this rule.
4	There are products that sell well during specifi seasons.	Nielsen IQ's 2023 report indicates tha certain products sell better in specifi seasons.

5	There will be fluctuations in the daily quantit of products sold.	The research team has observed an established this rule.
6	There will be fluctuations in sales of distin products and product categories.	The research team has observed an established this rule.

1.1.2. Data generation

Using appropriate simulation facilities informed by extensive research on 21st-century business practices, we propose a simulated café takeaway location. This Canadian café takeaway brand, namely Bubble, was established in March 2022 and officially opened on April 1st of the same year. Operating under a takeaway business model, this store offers a diverse range of 19 high-quality beverages, available in two sizes: M (medium) and L (large). Throughout the dataset period from 2022 to 2024, the store's management team consistently comprises three managers. The shop generally operates from 6:00 AM to 11:30 PM daily, except during special events.

The final database for the entire store consists of 11 tables, covering accounts, profiles, the menu, and order history. Accounts and profiles are automatically generated using two powerful data generation tools: Mockaroo and Generatedata. The menu is crafted using real-life examples of ingredients and products from popular brands like Starbucks and Highlands Coffee. Finally, the order history is created using Python libraries, mostly Pandas and Numpy.

1.1.3. Data Understanding

Among the two tables presenting order history, the "orderMasters" table is created first. This table contains the master data for all orders placed by customers within the dataset's time frame. Its primary key, "OrderID," will be referenced as a foreign key in the "orderDetails" table, where "OrderDetailID" uniquely identifies each product (a unique ProductID) ordered. The quantity of OrderDetailID an OrderID has is equal to the number of unique products involved. While the "orderDetails" table functions similarly to a common receipt, which is fully visible to related customers, the "orderMasters" table includes information accessible only to staff and managers. This structure accommodates retailers' preference for applying different discount values to various products within the same order.

Table 1.2: Table data OrderMasters

Columns	Data Type	Data Type in MQL	Meaning
OrderID (PK)	INT	VARCHAR(255)	The primary key of the table, uniquely identifying each order.
CustomerID	CHAR(6)	VARCHAR(255)	The identifier for the customer, specifying which customer placed the order.
Order_Date	DATE	TEXT	The date when the order was placed.
Order_Status	VARCHAR(10)	TEXT	The current status of the order.
EmployeeID	INT	TEXT	The identifier for the employee who processed the order.
ManagerID	INT	TEXT	The identifier for the manager overseeing the order process.
Order_TimeStart	TIME	TEXT	The start time of the order processing.
Order_TimeEnd	TIME	TEXT	The end time of the order processing.

Table 1.3: Table data OrderDetails

Columns	Data Type	Data Type in MQL	Meaning
OrderDetailID (PK)	INT	VARCHAR(255)	The primary key of th table, uniquel identifying each order

OrderID	INT	VARCHAR(255)	The primary key of th table, uniquel identifying each order
ProductID	CHAR(5)	VARCHAR(255)	The identifier for th products , specifyin which product is bein referred to .
Product_Qty	INT	TEXT	The quantity of produc ordered.
Review_Star	INT	TEXT	The customer rating fo each product, typicall ranging from 1 to 5.
PriceChange	INT	TEXT	The discour percentage on eac product for thi particular order detail.

Table 1.4: Table data Recipes

Columns	Data Type	Data Type in MQL	Meaning
ProductID	CHAR(5)	VARCHAR(255)	The identifier for the products, specifying which product is being referred to .
IngredientID	CHAR(3)	VARCHAR(255)	The identifier for the products, specifying which product is being referred to .
Ingredient_Qty	DECIMAL(5,2)	TEXT	The quantity of the ingredient used in each product.

Table 1.5: Table data Products

Columns	Data Type	Data Type in MQL	Meaning

ProductID (PK)	CHAR(4)	VARCHAR(255)	The primary key of the table, uniquely identifying each product.
ProductCategoryID	CHAR(3)	VARCHAR(255)	The unique identifier linking each product to its category.
Product_Name	VARCHAR(25)	TEXT	The name of the product.
Product_Size	CHAR(1)	TEXT	The size of the product.
Product_UnitPrice	INT	TEXT	The price of a single unit of the product.
Product_Availability	INT	TEXT	Indicates whether the product is available for purchase. Typically, ‘1’ means available, and ‘0’ means not available.
PriceChange	INT	TEXT	The discount percentage on each product for this particular order detail.

Table 1.6: Table data ProductCategories

Columns	Data Type	Data Type in MQL	Meaning
ProductCategoryID(PK)	CHAR(3)	VARCHAR(255)	The primary key of the table, uniquely

			identifying each product category.
ProductCategory_FullName	VARCHAR(25)	TEXT	The name of a product category.

Table 1.7: Table data Ingredients

Columns	Data Type	Data Type in MQL	Meaning
IngredientID (PK)	CHAR(3)	VARCHAR(255)	The primary key of the table, uniquely identifying each ingredient.
Ingredient_FullName	VARCHAR(25)	TEXT	The primary key of the table, uniquely identifying each ingredient.
Ingredient_UnitAmount	VARCHAR(10)	TEXT	The unit of measurement for the quantity to the ingredient.
Ingredient_UnitAmount	INT	TEXT	The quantity of the ingredient, typically measured in the unit specified in 'Ingredient_UnitAmount' column.
Ingredient_UnitAmount	CHAR(3)	TEXT	The unique identifier linking each ingredient to its category

Table 1.8:Table data IngredientCategories

Columns	Data Type	Data Type in MQL	Meaning
IngredientCategoryID (PK)	CHAR(3)	VARCHAR(255)	The primary key of the table, uniquely identifying each ingredient category.
IngredientCategory_ FullName	VARCHAR(25)	TEXT	The name of the ingredient category.

Table 1.9: Table data Employees

Columns	Data Type	Data Type in MQL	Meaning
EmployeeID (PK)	INT	VARCHAR(255)	The primary key of the table, uniquely identifying each employee.
Employee_FirstName	VARCHAR(25)	TEXT	The first name of the employee.
Employee_LastName	VARCHAR(25)	TEXT	The last name of the employee.
Employee_Title	VARCHAR(10)	TEXT	The job title of the employee within the organization.
Employee_Birthday	DATE	TEXT	The date of birth of the employee.
Employee_HireDate	DATE	TEXT	The date when the employee was hired
Employee_EndDate	DATE	TEXT	The date when the employment of the employee ended, if applicable.

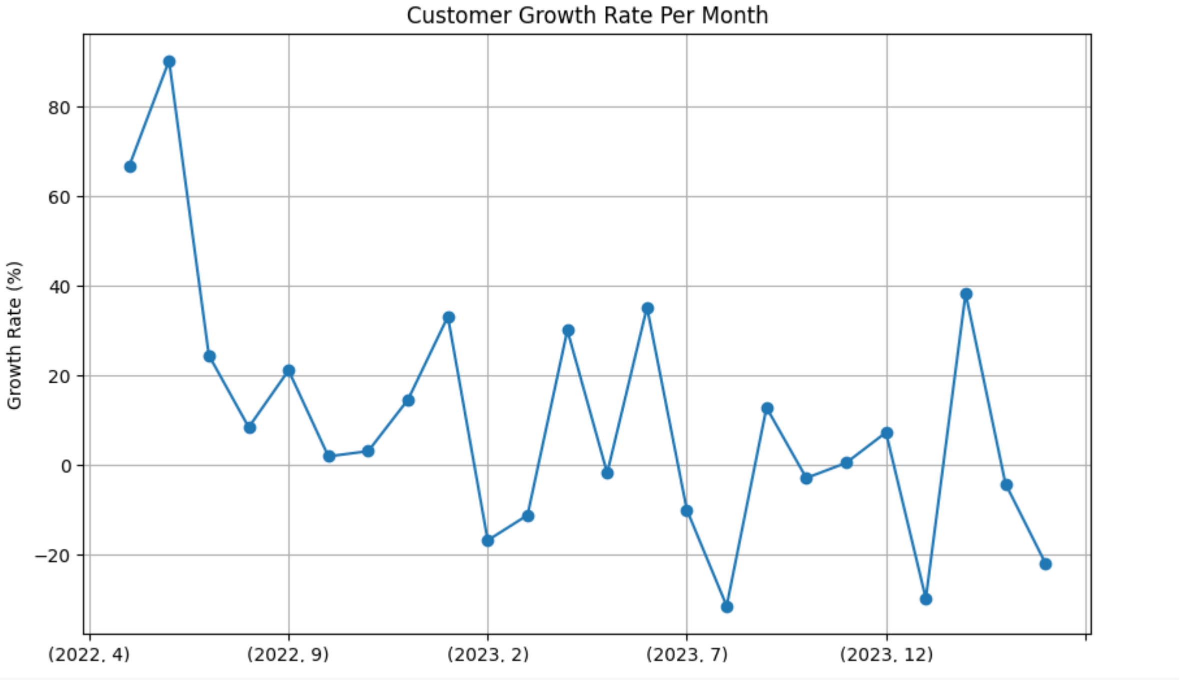
Table 1.10: Table data EmployeeAccs

Columns	Data Type	Data Type in MQL	Meaning
EmployeeID(PK)	INT	VARCHAR(255)	The primary key of the table, uniquely identifying each employee.
Employee_Username	VARCHAR(10)	TEXT	The username used by the employee for the login app.
Employee_Password	VARCHAR(25)	TEXT	The password associated with the employee's username for authentication.

1.1.4. Data Analysis:

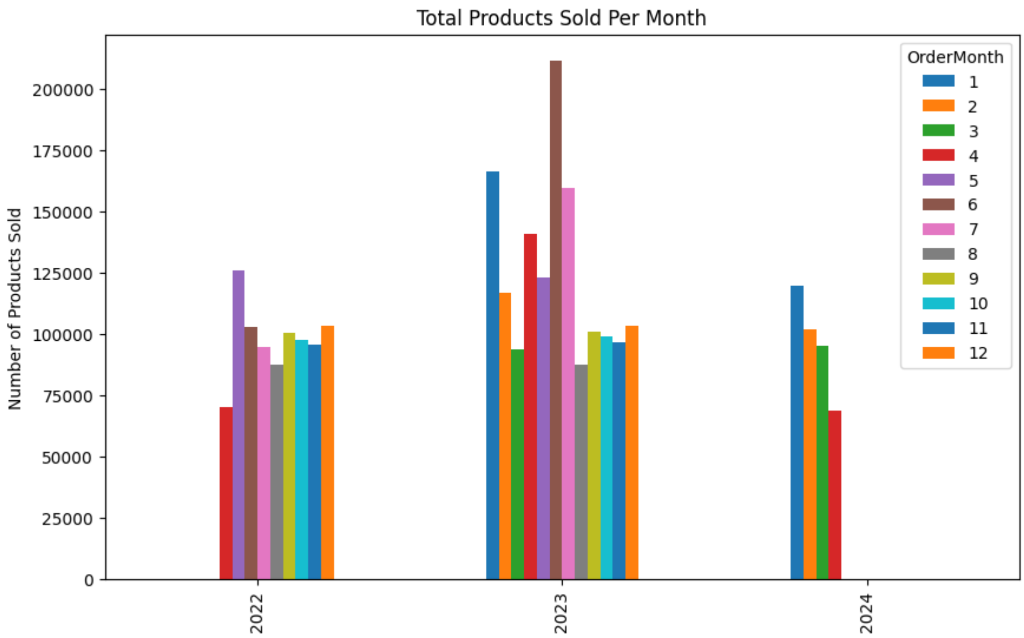
As of April 2024, the café has served 5,000 customers. The Customer Growth Rate per month, displayed below, shows significant fluctuations, indicating that the store has experienced its own challenging periods.

Figure 1.1: Customers' Growth Rate per month



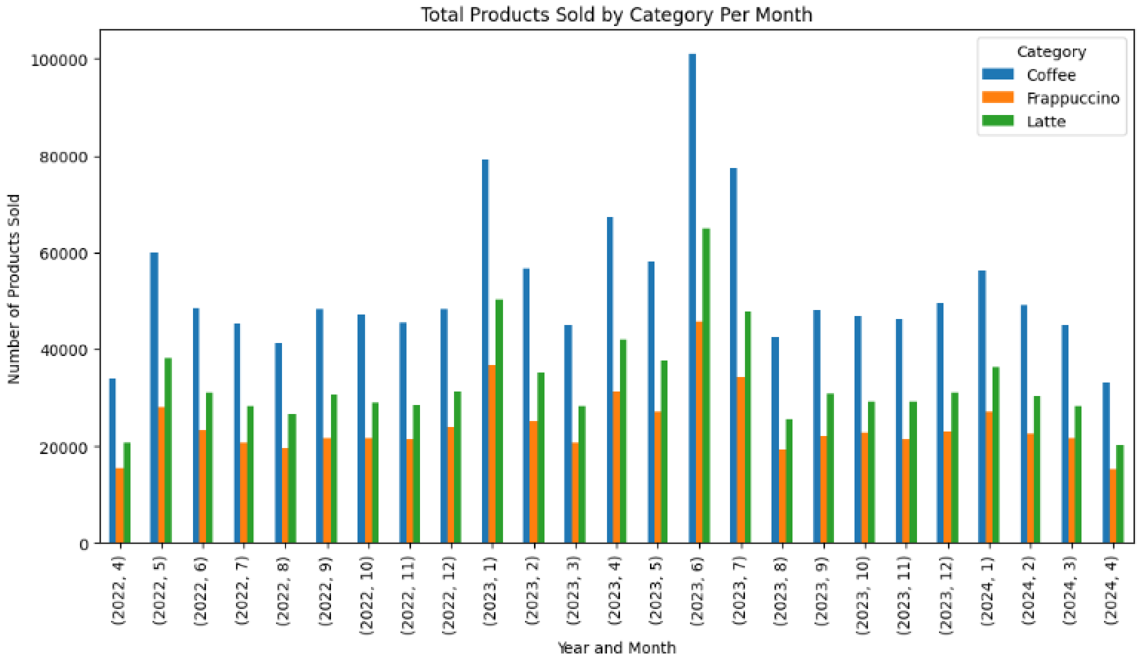
Special events in July 2023 led to a surge in sales. Additionally, the January New Year holiday earlier that year marked the second-highest number of products sold during the three years. In the first year, the number of products sold showed minimal fluctuation. During the normal months of the subsequent two years, the total number of products sold per month remained relatively consistent.

Figure 1.2: Number of Products sold per month



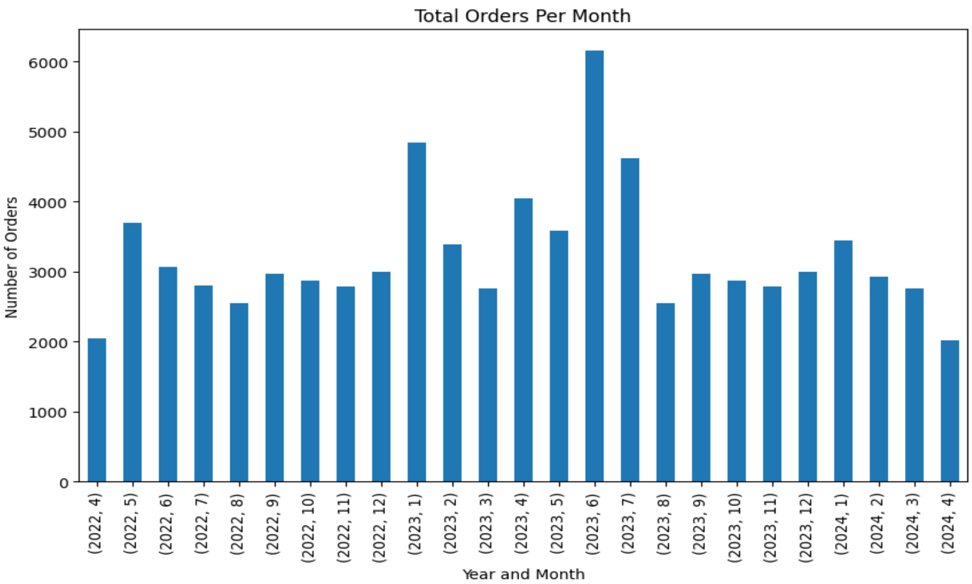
Customers have consistently preferred coffee over the two other categories of drinks. Overall, coffee remains the highest-selling product, followed by Frappuccino, with Latte closely following.

Figure 1.3: Number of Products sold by Product Category per month



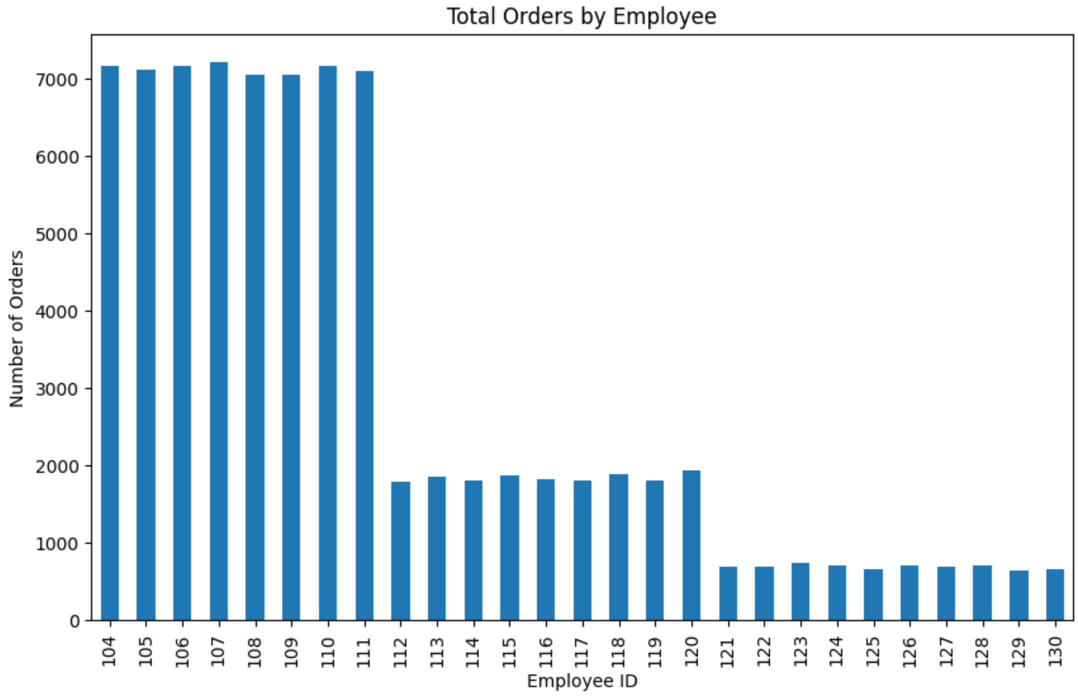
Generally, the store receives between 2,000 and 5,000 orders each month. The number of orders varies most significantly during holiday months.

Figure 1.4: Number of Products sold by Product Category per month



Among the 27 different Employee IDs ranging from 104 to 130, 8 staff members have handled a substantial number of orders, whereas 9 other employees have processed only around 2,000 orders, with the rest handling approximately 1,000 orders each.

Figure 1.5: Number of Orders received per Employee



Each order detail includes a Review Star rating. The average star rating varies notably between the top 20 customers and the least involved customers. The stars reviewed will help with Recommendation systems within the Customer Interface, ensuring customers with the best-suited choices introduced, giving them the best drinking experience.

Figure 1.6: Average star rating by the top 20 most satisfied Customers

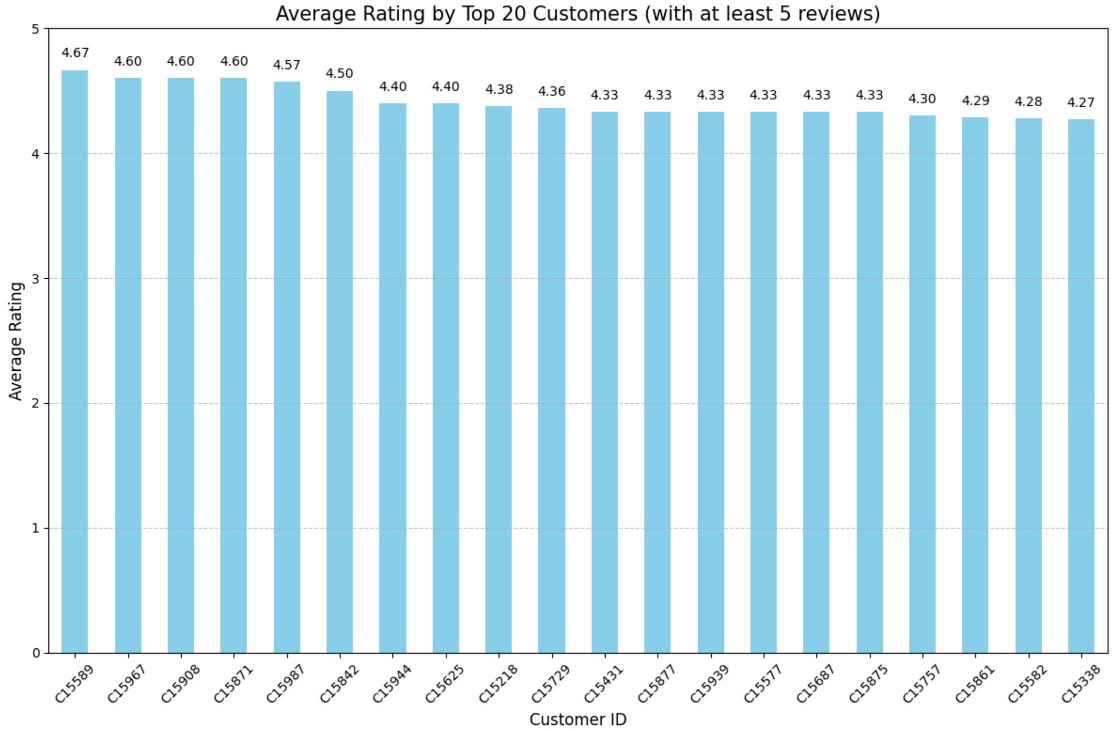
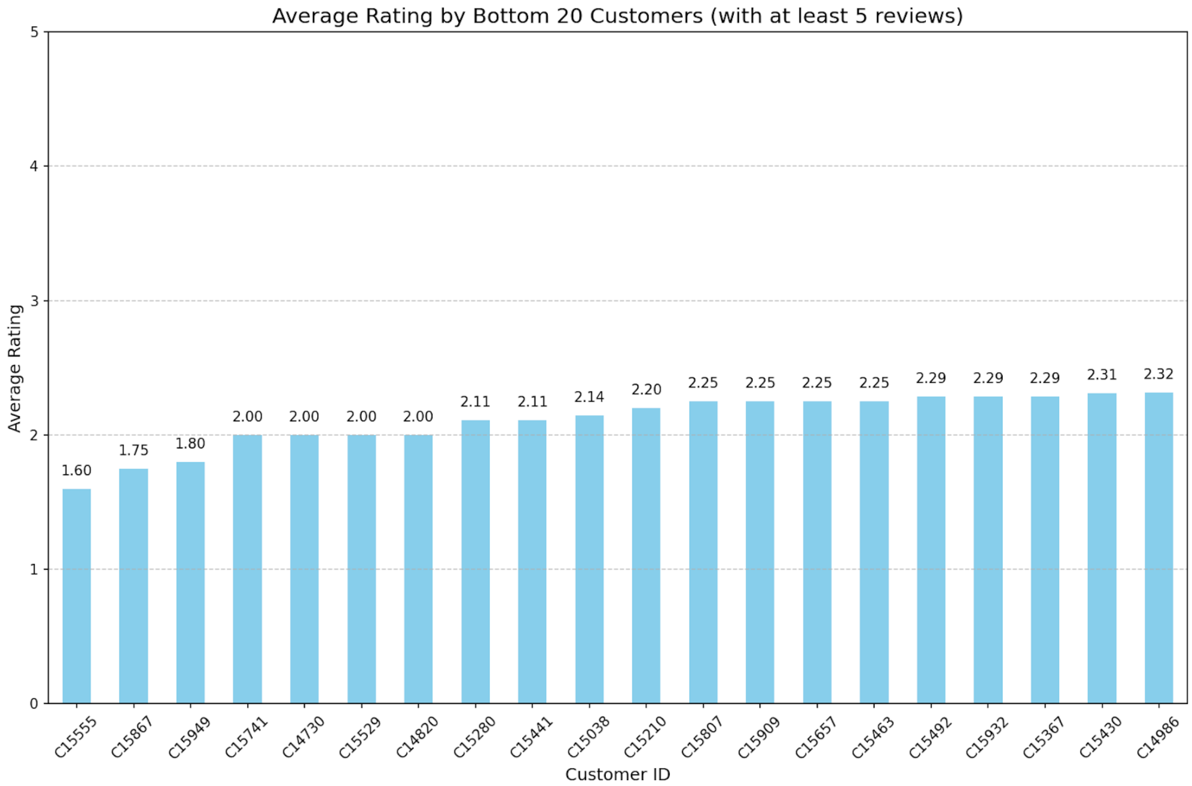


Figure 1.7: Average star rating by the top 20 most unsatisfied Customers



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Chapter 2. Methodology

This chapter outlines the methodologies employed in the research, detailing the programming languages, designing tools, and machine learning techniques used for sales forecasting.

2.1. Programming Languages

The programming languages chosen for this study are Python, specifically version 3.11. Python was selected due to its extensive libraries and frameworks such as NumPy, pandas, and sci-kit-learn, which facilitate data manipulation and machine learning tasks. Additionally, Python's simplicity and readability make it an excellent choice for developing and testing forecasting models as well as seamless functions.

2.2. Programming tools

Several design tools were utilized to streamline data analysis and visualization processes. Qt Design and PyQt6 were employed for creating the graphical user interface (GUI) of the applications, providing a powerful framework for designing and building interactive applications with rich user interfaces. Jupyter Notebooks, Kaggle, and Google Colab were used for data exploration and machine learning understanding, while PyCharm and Visual Studio Code were utilized for building interactive functions, training, and pickling models. This interactive environment facilitated exploratory data analysis and iterative model development. For visualization, libraries such as Matplotlib and Seaborn in Python were employed to create insightful graphs and charts, aiding in understanding data patterns and the performance of different models.

2.3. Machine Learning with Forecast

The machine learning techniques used in this study for sales forecasting include ARIMA, SARIMA, Exponential Smoothing, Random Forest, and XGBoost. Each method was chosen for its unique strengths in handling time series data and providing accurate predictions.

2.3.1. ARIMA

ARIMA (AutoRegressive Integrated Moving Average) is a classical time series forecasting method that combines autoregression, differencing, and moving averages. It is particularly effective for non-seasonal data. The ARIMA model was implemented using Python's statsmodels library, which offers a comprehensive suite of statistical tools for time series analysis. The model parameters were tuned using grid search to optimize the prediction accuracy.

2.3.2. SARIMA

SARIMA (Seasonal AutoRegressive Integrated Moving Average) extends ARIMA by including seasonal components, making it suitable for data with seasonality. The SARIMA model captures both the trend and seasonal patterns in the data, providing more accurate forecasts for seasonal items. The implementation was carried out using the same statsmodels library, with careful tuning of seasonal parameters to align with the periodic nature of the sales data

2.3.3. Exponential Smoothing

Exponential Smoothing techniques, including Simple, Holt's Linear, and Holt-Winters methods, were employed to forecast sales data. These methods smooth past observations to forecast future values, with Holt-Winters specifically designed to handle both trend and seasonality. The implementation utilized the statsmodels library in Python, which supports various exponential smoothing models. These models were particularly useful for their simplicity and effectiveness in handling different data patterns.

2.3.4. Random Forest

Random Forest, an ensemble learning method, was used for its robustness and ability to handle complex data structures. It constructs multiple decision trees during training and outputs the mean prediction of the individual trees, thus reducing overfitting and improving prediction accuracy. Python's scikit-learn library was used to implement the Random Forest model, leveraging its functionality to handle large datasets and perform feature importance analysis. This approach proved valuable in capturing the nonlinear relationships within the sales data.

2.3.5. XGBoost (Extreme Gradient Boosting)

XGBoost is an advanced implementation of gradient boosting designed for speed and performance. It is highly effective in handling large-scale data and complex patterns, making it a powerful tool for sales forecasting. XGBoost excels in managing missing data and providing high predictive accuracy, thanks to its ability to model intricate relationships in the data through an ensemble of weaker prediction models.

2.4. Machine Learning with Collaborative Recommendation

Collaborative recommendation is a widely used approach in machine learning, particularly in the domain of recommender systems. This method leverages the preferences of a large number of users to generate personalized recommendations. In this study, collaborative recommendation techniques were employed to enhance the accuracy and relevance of sales forecasts by analyzing user behavior and preferences.

User-Based Collaborative Filtering

User-Based Collaborative Filtering (UBCF) is a method that recommends items to a user based on the preferences of other users who have similar tastes. The process involves calculating the similarity between users using metrics such as Pearson correlation or cosine similarity. Once similar users are identified, their preferences are aggregated to generate recommendations.

Content-Based Filtering

Content-Based Filtering recommends items to users based on the attributes of items and the preferences or past behavior of the user. This method constructs a profile for each user or item, often utilizing keywords or features, and recommends items that match the user's profile. This approach is effective in recommending items that are similar to those the user has shown interest in previously.

Hybrid Approach

A hybrid approach combines content-based filtering with collaborative filtering to leverage the strengths of both methods. Content-based filtering recommends items based on item attributes and user preferences, while collaborative filtering relies on user interaction data. By integrating these techniques, the hybrid approach aims to improve recommendation accuracy and provide more relevant suggestions to users. This approach mitigates the limitations of each method when used in isolation, such as the cold-start problem in collaborative filtering and the limited scope of content-based filtering.

2.5 Evaluation Metrics

The Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) were employed as primary metrics for evaluating the performance of the forecasting models.

RMSE measures the square root of the average of squared differences between predicted and actual values, providing a comprehensive measure of forecast accuracy. Lower RMSE values indicate better model performance, and it penalizes larger errors more significantly.

MAE calculates the average of absolute differences between predicted and actual values, offering a straightforward measure of prediction accuracy. MAE is less sensitive to large errors compared to RMSE.

MAPE expresses the accuracy as a percentage, measuring the absolute error relative to actual values. It is particularly useful for understanding the forecast error in relative terms.

Chapter 3. Literature Review

3.1. Revenue Forecasting System

Sales forecasting is vital for business strategy, enabling companies to predict future demand based on environmental factors (Mentzer & Moon, 2004). Despite the inherent errors due to changing conditions, accurate forecasting is crucial. It's widely used across industries as an essential planning input. Key challenges include reducing prediction errors, often measured by metrics like Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). Advanced statistical methods and machine learning algorithms are frequently explored to improve accuracy by analyzing historical data and identifying patterns (Yasaman Ensafi et al., 2022). Ongoing advancements highlight the importance of predictive methods in maintaining competitiveness.

3.1.1. Classical Time-Series Forecasting Methods

No single method fits all time-series forecasting problems (Zhang & Kline, 2007). Moving Average (MA) is a simple method, especially for data without clear seasonal patterns (Chopra & Meindl, 2016). More advanced is the Autoregressive Integrated Moving Average (ARIMA). For instance, ARIMA and exponential smoothing were used to forecast women's footwear sales (Ramos, Santos, & Rebelo, 2015). Multivariate ARIMA has been applied to predict demand for perishable goods (Huber, Gossmann, & Stuckenschmidt, 2017). ARIMA combined with neural networks has been used for network traffic forecasting (Yang et al., 2021) and cryptocurrency price prediction based on seasonal ARIMA (SARIMA) has been used for applications like tourism demand (Goh & Law, 2002) and traffic flow (Williams & Hoel, 2003). However, SARIMA's linear nature limits its ability to detect nonlinear and highly volatile patterns (Hamzaçebi, 2008). In response, Seasonal Support Vector Regression (SSVR) was introduced, proving superior to SARIMA for seasonal items (Pai, Lin, Lin, & Chang, 2010). Exponential smoothing, including the Holt-Winters (HW) technique, is widely used for handling seasonality (Kalekar, 2004; Taylor, 2011). Advanced methods include models for retail store locations (Lv, Bai, Yin, & Dong, 2008) and Greedy Aggregation Decomposition (GAD) for managing fashion retailer inventories (Li & Lim, 2018), social media trends (Tandon, Revankar, & Parihar, 2021).

There are different types of exponential smoothing methods, with the Holt-Winters (HW) technique being particularly notable for its ability to address seasonality in datasets (Kalekar, 2004). Taylor (2011) applied a seasonal exponential smoothing method to forecast daily supermarket sales, successfully managing high volatility.

Some studies have utilized more advanced techniques. Lv, Bai, Yin, and Dong (2008) proposed a new model for evaluating the new locations of retail stores based on sales forecasting. Li and Lim (2018) proposed a Greedy Aggregation Decomposition (GAD) method to overcome the limited space in a fashion retailer's inventory related to Stock Keeping Units (SKU) daily demands. Additionally, Giering (2008) implemented a system for online product recommendations based on large retail store sales prediction.

3.1.2. Random Forest and XGBoost

Today, people use machine learning and powerful algorithms to address a variety of problems. By applying machine learning models such as XGBoost, Random Forest, and Support Vector Machines, it is possible to predict sales values more accurately and obtain refined results (Bohanec, Borštnar, & Robnik-Šikonja, 2017).

Random Forest, introduced by Liaw and Wiener (2002), is an ensemble learning method that constructs multiple decision trees and merges them to get a more accurate and stable prediction. This method has been widely applied across various domains due to its robustness and ability to handle large datasets. For example, Random Forest has been used to predict stock prices, where it was able to capture the complex patterns in financial data (Patel et al., 2015). In retail, it has been applied to forecast customer demand, helping businesses optimize inventory and reduce costs (Lian, 2016). Furthermore, Random Forest has been utilized in healthcare for predicting disease outbreaks, providing valuable insights for preventive measures (Cheng et al., 2014), and in agriculture for predicting crop yields, thereby aiding in better resource management (Belgiu & Drăguţ, 2016). Random Forests excel in reducing overfitting and improving the generalization of the model, offering a powerful tool for complex prediction tasks.

XGBoost, developed by Chen and Guestrin (2016), is a gradient-boosting framework that has gained widespread popularity for its superior performance in various machine-learning competitions and real-world applications. XGBoost uses a boosting technique that sequentially builds trees, where each new tree corrects errors made by the previous ones. This method is particularly effective for capturing intricate patterns and interactions within the data, leading to high prediction accuracy. XGBoost has been successfully applied in sales forecasting, demonstrating its ability to improve accuracy over traditional methods (Zhou et al., 2018). In the financial sector, XGBoost has been used for predicting loan defaults, providing banks with more accurate risk assessments (Nayak et al., 2019). Additionally, in the energy sector, it has been employed for load forecasting, helping utility companies manage demand more efficiently (Panapakidis & Dagoumas, 2017). In the field of credit scoring, XGBoost has been applied to develop more reliable credit models, thereby enhancing the decision-making process for lending institutions (Tianqi et al., 2015).

In summary, Random Forest and XGBoost represent state-of-the-art techniques in time-series forecasting. Their ability to handle large, complex datasets and improve prediction accuracy makes them valuable tools for maintaining competitiveness and driving strategic business decisions. By leveraging these advanced machine learning methods, businesses can achieve more reliable forecasts, optimize operations, and enhance overall performance.

3.1.3. Prophet and Artificial Neural Networks for Time-Series Forecasting

The Prophet method, developed by Facebook (Taylor & Letham, 2018), is relatively new. Studies show Prophet's results comparable to traditional methods (Papacharalampous, Tyralis, & Koutsoyiannis, 2018) and more accurate than SARIMA for hospital discharge predictions (McCoy, Pellegrini, & Perlis, 2018). Prophet has also been used for predicting website traffic (Subashini, Sandhiya, Saranya, & Harsha, 2019) and COVID-19 cases in Indonesia (Satrio, Darmawan, Nadia, & Hanafiah, 2021).

Artificial Neural Networks (ANN) offer another powerful forecasting approach. ANN outperformed traditional methods like Winters' exponential smoothing and Box Jenkins for predicting aggregate retail sales (Alon, Qi, & Sadowski, 2001). ANN's ability to identify patterns in data makes it popular for sales forecasting (Tkáč & Verner, 2016). The importance of data pre-processing for accurate predictions with Multilayer Perceptrons (MLP) has been highlighted (Zhang & Qi, 2005; Sharma, Rana, & Kumar, 2021). Enhancements like the Seasonal Artificial Neural Network (SANN) (Hamzaçebi, 2008) and hybrid ANN-ARIMA models (Khashei & Bijari, 2010) have improved prediction accuracy.

3.1.4. Deep learning techniques

Deep learning techniques are essential for accurate forecasting. Reviews have explored Swarm intelligence algorithms (Kotsiantis, Zaharakis, & Pintelas, 2006), bio-inspired algorithms (Kar, 2016; Chakraborty & Kar, 2017), and big data applications (Kar & Dwivedi, 2020).

Deep neural networks, such as Multilayer Perceptrons (MLP), can model complex functions (Alpaydin, 2009). They have been used for solar power forecasting (Gensler et al., 2016), stock price prediction (Zhuge et al., 2017), and traffic forecasting (Zhao et al., 2017). Deep learning models for retail sales prediction (Kaneko & Yada, 2016) and product ranking based on online reviews (Lee, Rim, & Lee, 2019) have shown superior results.

Recurrent Neural Networks (RNNs), especially Long Short-Term Memory (LSTM) networks, are useful for time-series prediction due to their ability to retain past events (Gamboa, 2017). LSTM has been used for anomaly detection (Malhotra et al., 2016), forecasting (Zhao et al., 2017), and predicting market fear during COVID-19 (Ghosh & Sanyal, 2021), although machine learning models like XGBoost sometimes perform better.

Convolutional Neural Networks (CNNs) have also been used for time-series forecasting. CNNs have been applied to sales forecasting (Zhao & Wang, 2017), stock price forecasting (Selvin et al., 2017), and short-term solar power and electricity forecasting (Koprinska, Wu, & Wang, 2018). A robust CNN-based forecasting method was developed by (Momeny et al., 2021).

In summary, deep learning techniques like MLP, RNN, LSTM, and CNN are highly effective for various forecasting applications, offering significant advantages over traditional methods.

3.2. Products Recommendation System

Recommendation systems have become a significant area of research due to their increasing relevance in various industries such as e-commerce, entertainment, and social media. Researchers have explored various techniques including collaborative filtering, content-based filtering, hybrid methods, and machine learning algorithms to enhance recommendation accuracy and efficiency.

3.2.1. Machine Learning in Recommendation Systems

The integration of machine learning techniques has significantly enhanced the performance of recommendation systems. For instance, "Development of a Recommendation System Using Machine Learning Algorithms" by Zhang Wei, Li Ming, and Chen Qiang (2022) describes the development of a sophisticated recommendation system using machine learning algorithms. Their study demonstrates the practical application of neural networks and deep learning models in predicting user preferences, highlighting the improved accuracy over traditional methods. This research also delves into the challenges of overfitting and the importance of cross-validation techniques to ensure robust model performance.

Complementing this, "Building Recommender Systems with Advanced Algorithms" by Gupta Prashant, Kumar Saurav, and Singh Ankit (2023) provides a comprehensive guide to building recommender systems using advanced algorithms. Their work covers a variety of techniques

essential for managing large datasets, such as gradient boosting, ensemble methods, and hyperparameter tuning. They emphasize the importance of feature engineering and data preprocessing in enhancing the predictive power of machine learning models.
Additionally, "Utilizing Review Data for Building a Recommendation System" by Kumar Ramesh, Singh Pooja, and Sharma Divya (2021) utilizes review data to build a recommendation system, detailing the processes of data preprocessing, model building, and evaluation. Their research focuses on the use of sentiment analysis and natural language processing (NLP) to extract valuable insights from textual data, demonstrating how these methods can improve the accuracy of recommendations.

3.2.2. Collaborative Filtering

Collaborative filtering is one of the most popular techniques in recommendation systems, leveraging user-item interactions to make predictions. "Collaborative Filtering Techniques and Their Applications" by Park Jinwoo, Choi Yuna, and Lee Sungmin (2018) offers a foundational understanding of this method, explaining its basic types and mechanisms. They discuss user-based and item-based collaborative filtering, illustrating implementation using user interaction matrices and similarity calculations. Their research underscores the efficiency of matrix factorization techniques such as singular value decomposition (SVD) in handling large datasets and improving recommendation accuracy.
Extending this discussion, "A Simple Beer Recommendation System Using Collaborative Filtering" by Lee Jihoon and Kim Daewon (2020) presents a simple beer recommendation system using collaborative filtering. Their study focuses on the step-by-step process from data preparation to model evaluation, providing practical insights into the implementation of collaborative filtering algorithms. They demonstrate how to handle data sparsity and cold start problems, common challenges in collaborative filtering.

3.2.3. Content-Based Filtering

Content-based filtering involves recommending items based on the similarity of item content to the user’s preferences. It requires a detailed understanding of item features and user profiles. "Advancements in Content-Based Filtering for Recommendations" by Smith John and Doe Richard (2023) reviews the latest advancements in content-based filtering, emphasizing the integration of contextual information to refine recommendations. Their research explores the use of feature extraction techniques such as TF-IDF and word embeddings in text-based recommendations. They highlight the effectiveness of content-based methods in domains like media and literature where user interaction data may be sparse or unavailable.

3.2.4. Hybrid Approaches and Contextual Information

Combining multiple recommendation techniques can yield better results than using a single approach. "Enhancing Machine Learning Algorithms in Recommendation Systems" by Brown Thomas and Smith Andrew (2021) explores methods to enhance machine learning algorithms in recommendation systems, emphasizing the integration of new technologies and data sources. They discuss the benefits of hybrid models that combine collaborative filtering and content-based methods, leveraging the strengths of each approach to improve recommendation quality. Their research includes case studies on e-commerce platforms, demonstrating how hybrid models can provide more personalized and accurate recommendations.

Additionally, "Consensus-Based Approaches in Hybrid Recommendation Systems" by Khan Waqas Ali and Ahmed Mohsin (2021) highlights the value of integrating contextual information into recommendation systems. Their study shows how incorporating factors such as time, location, and social context can significantly enhance the relevance of recommendations. They present a hierarchical model that adapts to changing user preferences over time, illustrating the dynamic nature of modern recommendation systems.

3.2.5. Specific Applications

Several studies explore the application of recommendation systems in niche areas. "Developing a Food Recommendation System Using Machine Learning" by Wang Hong and Zhang Lei (2021) develops a food recommendation system using collaborative filtering and machine learning techniques to cater to specific user preferences. Their system utilizes dietary restrictions and nutritional information to provide personalized meal recommendations. This application showcases the adaptability of recommendation systems to different user needs and highlights their potential in the health and wellness sectors.

3.2.6. Evaluation and Metrics

Accurate evaluation of recommendation systems is crucial for assessing their performance. "Enhancements to Collaborative Filtering Algorithms" by Doe Jane and Green Brian (2022) proposes enhancements to collaborative filtering algorithms to improve accuracy and efficiency, supported by experimental results. They discuss the use of evaluation metrics such as precision, recall, F1-score, and Mean Average Precision (MAP) to measure the performance of these systems. Their research emphasizes the importance of continuous model evaluation and validation to ensure that recommendation systems meet desired standards of accuracy and reliability.

3.2.7. Emerging Trends and Future Directions

The field of recommendation systems is continuously evolving, with new techniques and technologies emerging. "A Hierarchical Integrated Machine Learning Model to Predict Flight Departure Delays and Durations" by Khan Waqas Ali and Ahmed Mohsin (2021) presents a hierarchical integrated machine learning model to predict flight departure delays and durations, demonstrating the effectiveness of various algorithms and sampling techniques in achieving better accuracy. This study indicates the potential for further research and development in hierarchical models and integrated approaches. They discuss the use of advanced techniques such as reinforcement learning and multi-task learning to further enhance recommendation systems.

As new data sources and technologies become available, recommendation systems will continue to improve, offering more accurate and personalized recommendations. Future research directions include the exploration of deep learning models, real-time data processing, and the integration of multi-modal data to further refine recommendation algorithms.

Overall, the literature underscores the diverse approaches and continuous advancements in recommendation systems, highlighting the crucial role of machine learning and collaborative filtering in improving recommendation accuracy and user satisfaction. The integration of various techniques and the ongoing evolution of these systems reflect their growing importance and potential in a wide range of applications.

Chapter 4. Data preparation and Model building

4.1. Forecast

4.1.1. Import the dataset

The combined dataset covers sales data from the Bubble store for the period between June 2, 2022, and July 31, 2023. This dataset was created by merging data from three other sources:

OrderDetails: Contains information about specific products included in each order in the Bubble store.

OrderMasters: Contains information about how and when each order was created and the responsible parties.

Products: Contains information about the products offered by the Bubble store.

After merging these datasets, the combined dataset consists of 301,975 rows and 10 variables.

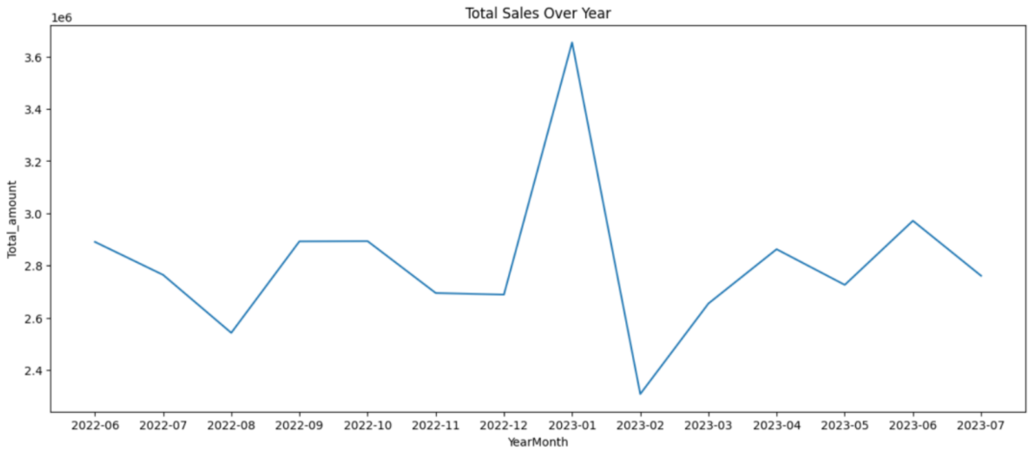
Table 4.1: Variables used

Variables	Describe
OrderID	Identifier of the order (detail)
ProductID	Product in the order (detail)
Product_Qty	Quantity of each product in the order (detail)
Product_Size	Size of the product (L and M)
Product_UnitPrice	Price of the product
PriceChange	Discount value
Order_Date	The date in which order (detail) was created
Order_TimeStart	Start time of an order (detail)
Order_TimeEnd	End time of an order (detail)
Total_amount	Total amount paid by the customer

4.1.2. Inspect the data

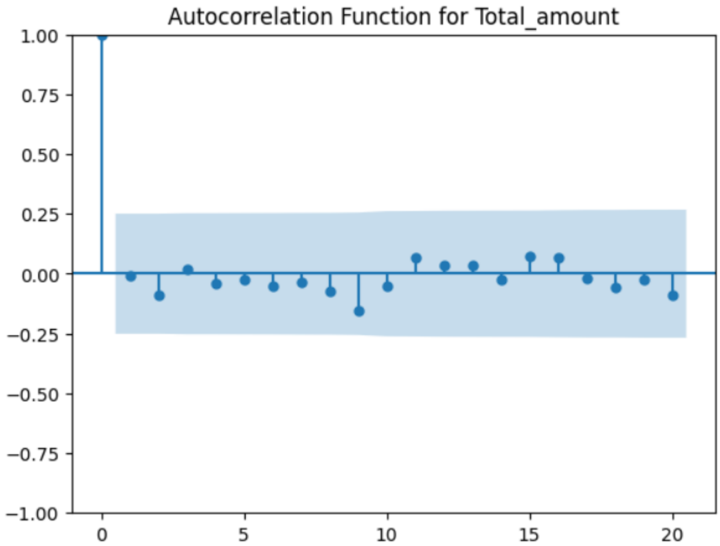
After having the required dataset, we start to implement explanatory analysis. First, we generate a line chart to observe the fluctuation of total sales over the year.

Figure 4.1: Total sales over year



From June to November 2022, total sales showed little fluctuation, despite some periods of decline. A significant change occurred between December 2022 and January 2023, with total sales increasing dramatically before sharply decreasing again in February 2023. Following this, the total sales trend showed a steady increase from March 2023 to July 2023.

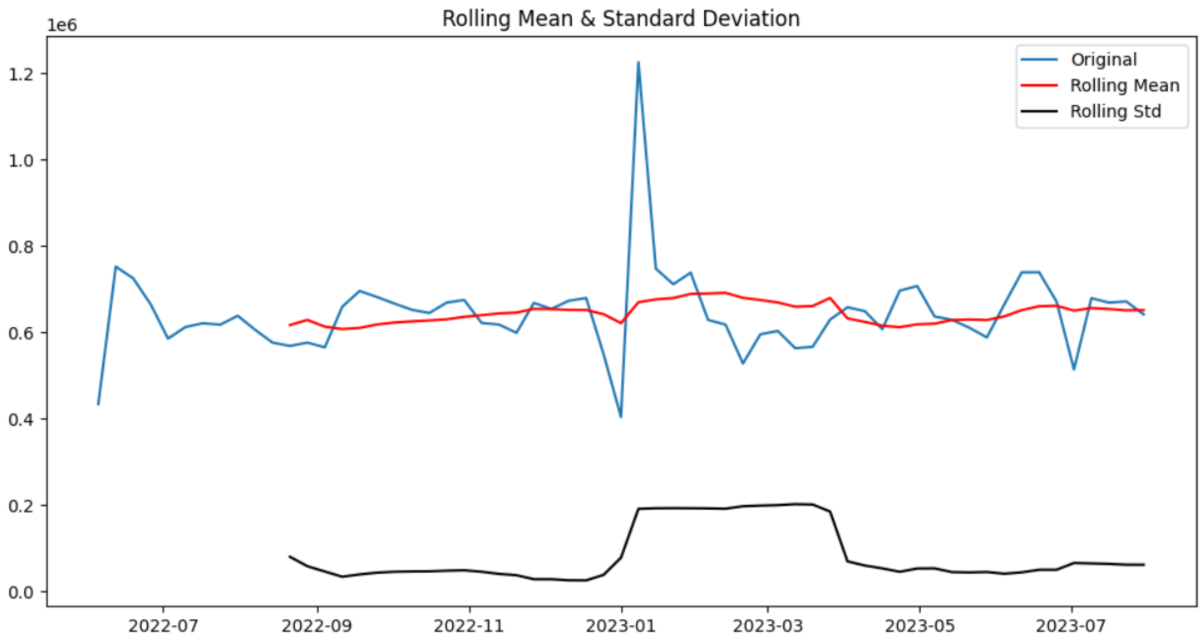
Figure 4.2: Autocorrelation function for total amount



The Autocorrelation Function (ACF) measures the correlation between observations in a time series at different time lags. At lag 0, the autocorrelation is always 1 because a time series is perfectly correlated with itself at this lag. For other lags, the autocorrelation values should ideally fall within the blue-shaded confidence interval (usually a 95% confidence interval). If they fall outside this interval, it indicates a significant correlation at those lags.

In the plot, none of the autocorrelation values for lags beyond 0 significantly exceed the blue-shaded area, indicating no significant autocorrelation at these lags. A stationary time series typically has autocorrelations that diminish quickly to zero. The ACF plot shows that most autocorrelations for lags beyond 0 are close to zero and within the confidence interval.

Figure 4.3: Rolling Mean and Standard Deviation



Original Sales Data (Blue Line)

The sales data exhibits moderate fluctuations but remains relatively stable during the period from June to November 2022. A notable spike in sales occurs around December 2022, followed by a sharp decline in January 2023. After this decline, sales started to increase again from March 2023, showing some variability but an overall upward trend.

Rolling Mean (Red Line)

The rolling mean shows a gradual increase from June to November 2022, smoothing out short-term fluctuations and reflecting the overall trend. Despite the significant spike in December 2022, the rolling mean remained relatively steady during the December 2022 to January 2023 period, suggesting the spike was a short-term anomaly. From March 2023 onwards, the rolling mean starts to increase more noticeably, indicating a consistent upward trend in sales.

Rolling Standard Deviation (Black Line)

The rolling standard deviation is low and stable from June to November 2022, indicating consistent sales without much variability. There is a sharp increase in the rolling standard deviation around December 2022, corresponding to the spike in sales, followed by a sharp decrease in January 2023. After February 2023, the rolling standard deviation decreases and remains low, indicating more consistent sales with less variability.

The rolling mean indicates that the average sales level changes over time, particularly with a noticeable increase from March 2023 onwards. In addition, the rolling standard deviation shows variability, especially around the December 2022 to January 2023 period while it stabilizes somewhat after February 2023, the earlier variability indicates that the variance is not constant over time. The presence of a changing mean and periods of fluctuating variability suggest non-stationarity.

ADF and KPSS test

The ADF test yields a statistic of -7.96 with a p-value critical low which approximates 0. This strongly indicates that the time series is stationary, as the statistic is significantly lower than the critical values at the 1%, 5%, and 10% levels. Completely, the KPSS test produces a statistic of 0.099 with a p-value of 0.1, suggesting that we cannot reject the null hypothesis of stationarity. Together, these results confirm that the time series is stationary.

Although the visual data suggests non-stationarity, the ADF and KPSS tests show strong statistical evidence of stationarity in the time series. While both visual and statistical analyses are important, the latter provides a more reliable conclusion here. Rolling statistics are sensitive to short-term fluctuations and anomalies, the ADF and KPSS tests are designed to evaluate broader trends and are less influenced by temporary deviations. Visual analysis using rolling statistics, like mean and standard deviation, can be influenced by short-term fluctuations in the data. These fluctuations might create the illusion of non-stationarity, even if the underlying process is stable. The rolling mean and standard deviation capture changes in the data over small windows, highlighting local variations that don't necessarily affect the overall stationarity of the series. Anomalies, such as a spike in sales around December 2022 followed by a decline in January 2023, are significant but may not substantially alter the overall properties of the time series as assessed by the ADF and KPSS tests.

4.1.3. Preprocess the data

Initially, anomalies were removed from the time series. Following this, unnecessary columns were removed, and the data was resampled to compute the total sales amount per day. These totals were then used as input for ARIMA, SARIMA, Exponential Smoothing, Random Forest, and XGBoost models. For Random Forest and XGBoost, the input variables comprised the mean of Product_UnitPrice and the sum of Product_Qty sales for each day.

4.1.4. Model training and validation

After preprocessing, the data will be divided into two sets: a training set and a test set, with the training set comprising 80% of the data. We then customized the input data for the models used in this study, including ARIMA, SARIMA, Exponential Smoothing, Random Forest, and XGBoost. The performance of the trained models will be evaluated using metrics such as RMSE (Root Mean Square Error), MAE (Mean Absolute Error), and MAPE (Mean Absolute Percentage Error). The training results will be visualized below.

Figure 4.4: Training result of ARIMA

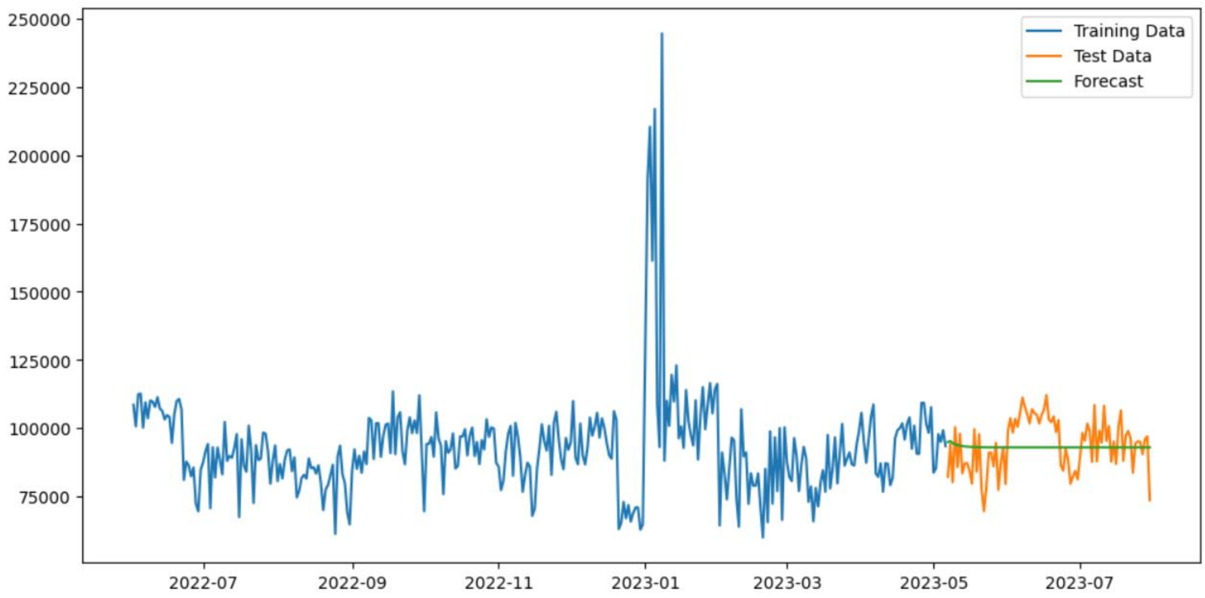


Figure 4.5: Training result of Exponential Smoothing

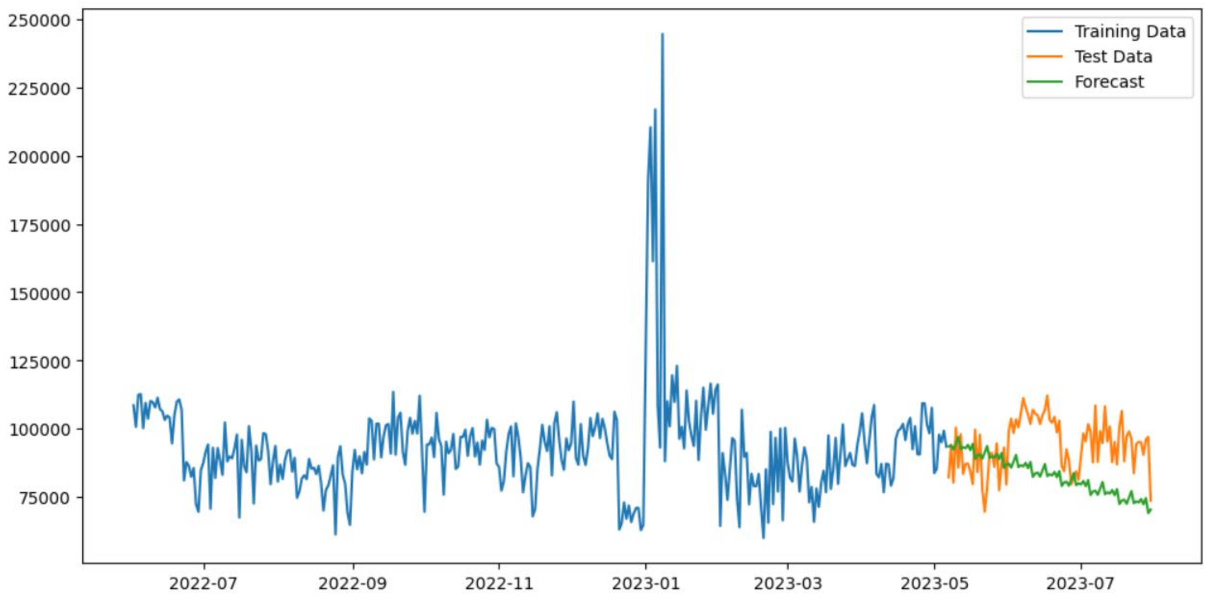
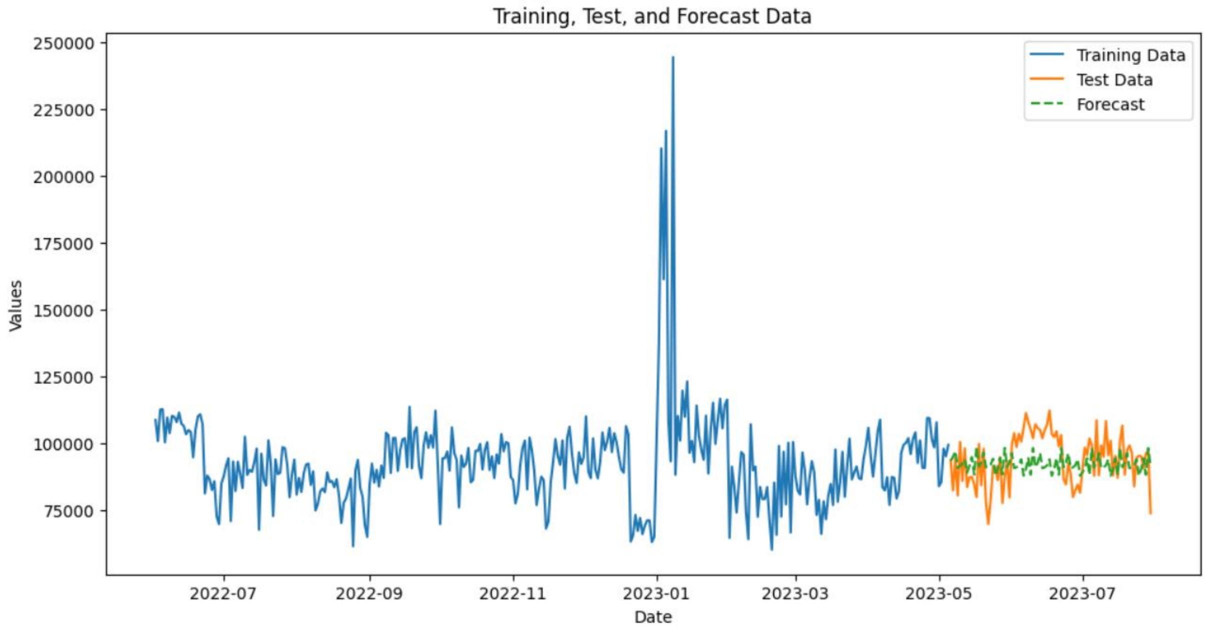


Figure 4.6: Training result of SARIMA



The visualization charts for time-series models depict three components: the blue line for training data, the orange line for test data, and the green dashed line for forecasting results. In the ARIMA chart, the forecast line initially dips slightly before leveling off. This indicates that the ARIMA model is predicting a stable value without accounting for the fluctuations or trends present in the test data. Consequently, the forecast fails to capture the variability observed in the test data, suggesting the model does not adequately reflect the underlying patterns of the time series. Regarding the Exponential Smoothing model, the forecast shows a downward trend during the test period. This suggests that the model anticipates a decrease over time, in contrast to the flat forecast produced by the ARIMA model. While the Exponential Smoothing model captures some degree of trend, it does not fully align with the variations seen in the test data. The SARIMA model, on the other hand, shows a forecast that closely aligns with the test data compared to both the ARIMA and Exponential Smoothing models. It captures some variability, indicating its ability to handle seasonal patterns and trends more effectively. Although the significant spike in early 2023 within the training data might influence the model's performance, the SARIMA model seems to manage it better by incorporating seasonal components.

Figure 4.7: Training result of Random Forest

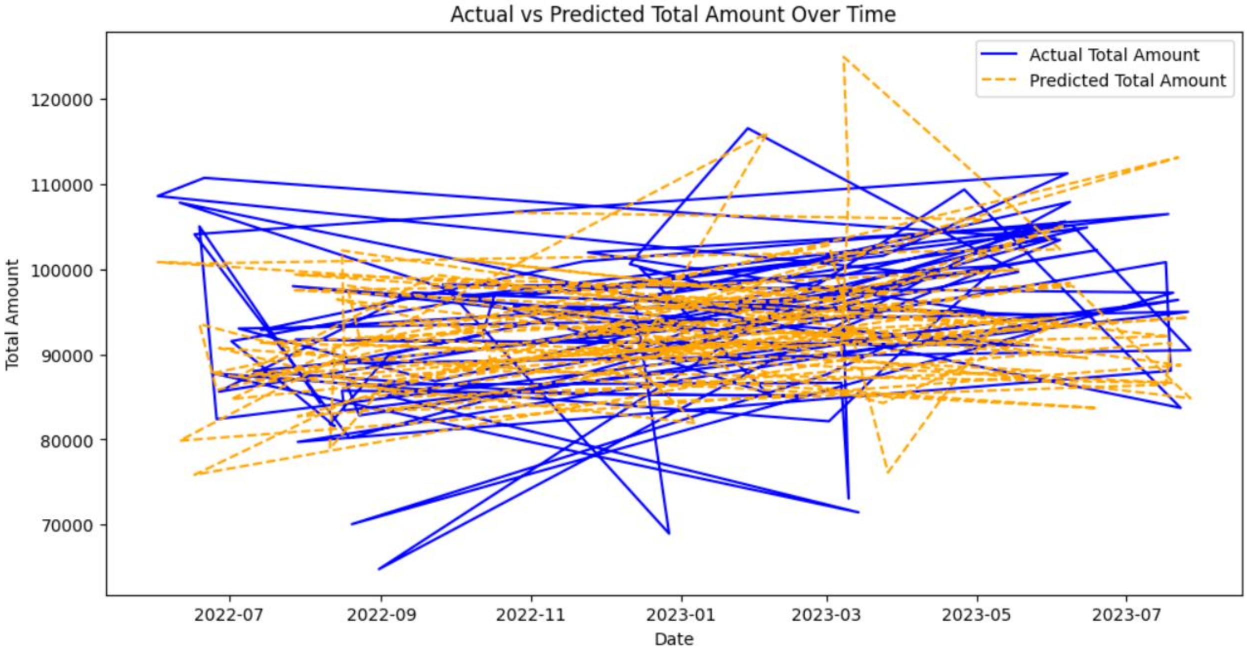
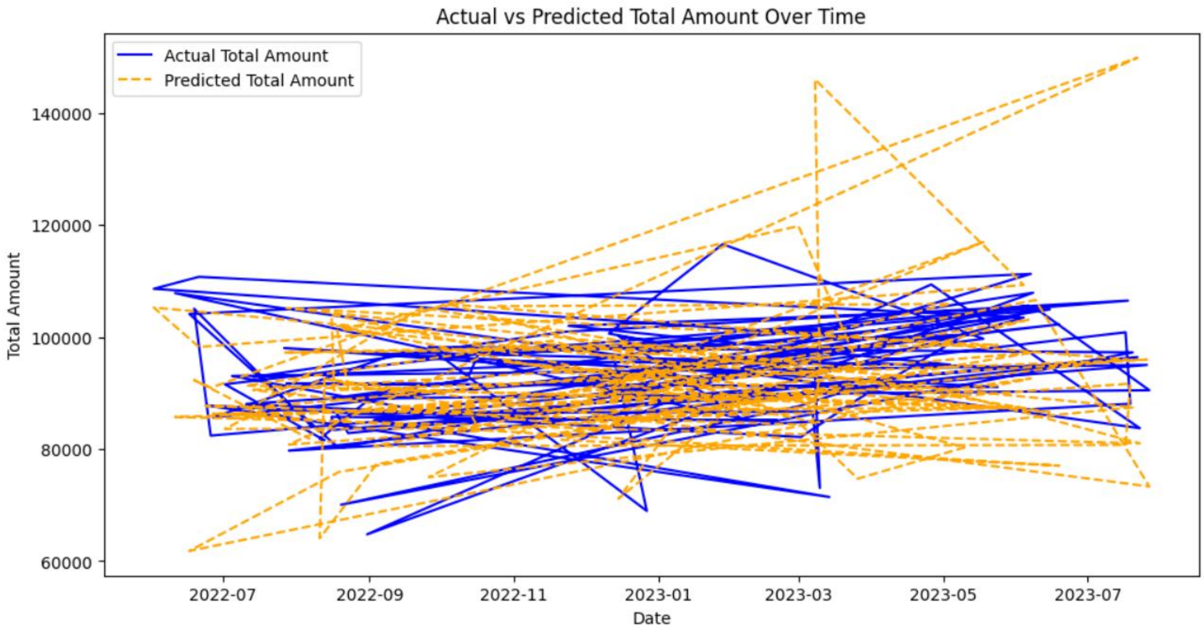


Figure 4.8: Training result of XGBoost



The visualization chart compares the actual total amounts (represented by the blue lines) with the predicted total amounts from Random Forest and XGBoost (represented by the orange dashed lines) over a period from mid-2022 to mid-2023. Both models exhibit a high degree of overlap and complexity, with numerous crossing lines that highlight the significant fluctuations and variability in the actual amounts over time.For the Random Forest model, the predicted total amounts show notable variability, yet they closely follow the general trend of the actual data. This indicates that Random Forest captures the overall trend quite well, though it struggles with short-term volatility.In comparison, the XGBoost model also aligns closely with the actual amounts but not as effectively as the Random Forest model. While XGBoost captures the general trend, it exhibits significant deviations, particularly during periods of abrupt changes in the actual data.

Overall, both models capture the overarching trend in the data, but Random Forest appears to handle the variability and trend alignment slightly better than XGBoost, which shows more pronounced deviations during sudden changes.

Table 4.2: Table of Evaluation metrics for models

Metrics / Models	RMSE	MAE	MAPE
ARIMA	9692.34	8323.61	9%
Exponential Smoothing	17050.19	14732.0	15.2%
SARIMA	9957.268	8311.25	8.9%
Random Forest	12654.56	9732.08	10.7%
XGBoost	16083.66	12006.8	13.08%

In order to find the best model, we first evaluate based on evaluation metrics including RMSE, MAE and MAPE. After comparing those metrics, Exponential Smoothing emerged as the least effective model, yielding RMSE, MAE, and MAPE values of 17050.19, 14732, and 15.2%, respectively. Following Exponential Smoothing, XGBoost exhibited slightly better performance with metrics results of 16083.66, 12006.8, and 13.08%. Random Forest surpassed both, achieving RMSE and MAE values of 12654.56 and 9732.08, respectively, with a MAPE of 10.7%. The models with the most promising results, ARIMA and SARIMA, were compared. SARIMA showed a higher RMSE (9957.268 > 9692.34) but lower MAE (8311.25 < 8323.61) and MAPE (8.9% < 9%) compared to ARIMA. Despite the marginal differences in performance metrics, SARIMA demonstrated a closer fit between the forecast and test data lines upon visual inspection. Therefore, SARIMA was chosen as the preferred model for application.

4.2. Recommender system

4.2.1. Import the dataset

The dataset we use to create the recommender system is the same as the datasets used to create forecasting models including OrderMasters, OrderDetails, and Products for the period between June 2, 2022, and July 31, 2023.

4.2.3. Preprocess the data

To implement content-based filtering, we will enhance the product dataset by creating a new column named ‘Content’. This column will combine information from the following five columns: ProductID, ProductCategoryID, Product_Name, Product_Size, and Product_UnitPrice. The combined data in the ‘Content’ column will then be vectorized using a TF-IDF vectorizer. For collaborative filtering, we will merge the Order Master and Order Details datasets. This merged dataset will contain three features: CustomerID, ProductID, and Review_Star.

4.2.4. Model training and validation

Content-based Filtering

First, we calculate the cosine similarity scores for the products using the vectorized values in the ‘Content’ column. We then create a method with two parameters: product_id and top_n. The product_id parameter specifies the ID of a product, and top_n is an integer that defines the number of products to recommend. Upon receiving the product ID, we obtain the similarity scores for the specified product. These similarity scores are then sorted in descending order. The indices of the top_n products are selected in descending order, excluding the first one as it likely corresponds to the product itself. Finally, the top_n products are returned as recommendations.

Collaborative Filtering

The dataset for collaborative filtering is loaded into the Surprise Dataset and then split into training and testing sets using the ‘surprise’ library. The training data is then fitted to SVD (Singular Value Decomposition) and KNN (K-Nearest Neighbors) models. The evaluation metrics used are RMSE (Root Mean Square Error) and MAE (Mean Absolute Error), and we implement k-fold cross-validation with 5 folds. The results of both models are presented below.

Table 4.3: Validations of KNN

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean	Std
RMSE	1.3327	1.3333	1.3311	1.3311	1.3388	1.3334	0.0028
MAE	1.1028	1.1073	1.1024	1.1041	1.1099	1.1053	0.0029
Fit time (minutes)	127.14	128.54	127.45	129.47	129.47	129.47	0.84
Test time (minutes)	527.67	534.77	523.65	519.63	534.75	528.10	6.01

Table 4.4: Validations of SVD

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean	Std
RMSE	1.3053	1.3082	1.3057	1.3055	1.3095	1.3069	0.001
MAE	1.0842	1.0872	1.0854	1.0851	1.0879	1.0860	0.001
Fit time (minutes)	3.56	3.72	3.64	3.61	3.69	3.64	0.06
Test time (minutes)	0.68	0.37	0.64	0.64	0.64	0.59	0.11

The average RMSE and MAE of the SVD model are slightly better than those of the KNN model, with RMSE values of 1.3069 compared to 1.3334, and MAE values of 1.0860 compared to 1.1053. However, the SVD model significantly outperforms KNN in terms of average fit and test times, with times of 3.64 seconds versus 129.47 seconds for fitting, and 0.59 seconds versus 528.10 seconds for testing. Therefore, we choose the SVD model for collaborative filtering.

Hybrid approach

After identifying the optimal methods and models for both content-based and collaborative filtering, we combine the recommendations from each approach to achieve the best filtering results. We first obtain the recommendation results from both methods and then merge these results into a single list. Duplicates are removed to ensure each recommended product is unique. Finally, we select the top products from this consolidated list for recommendation.

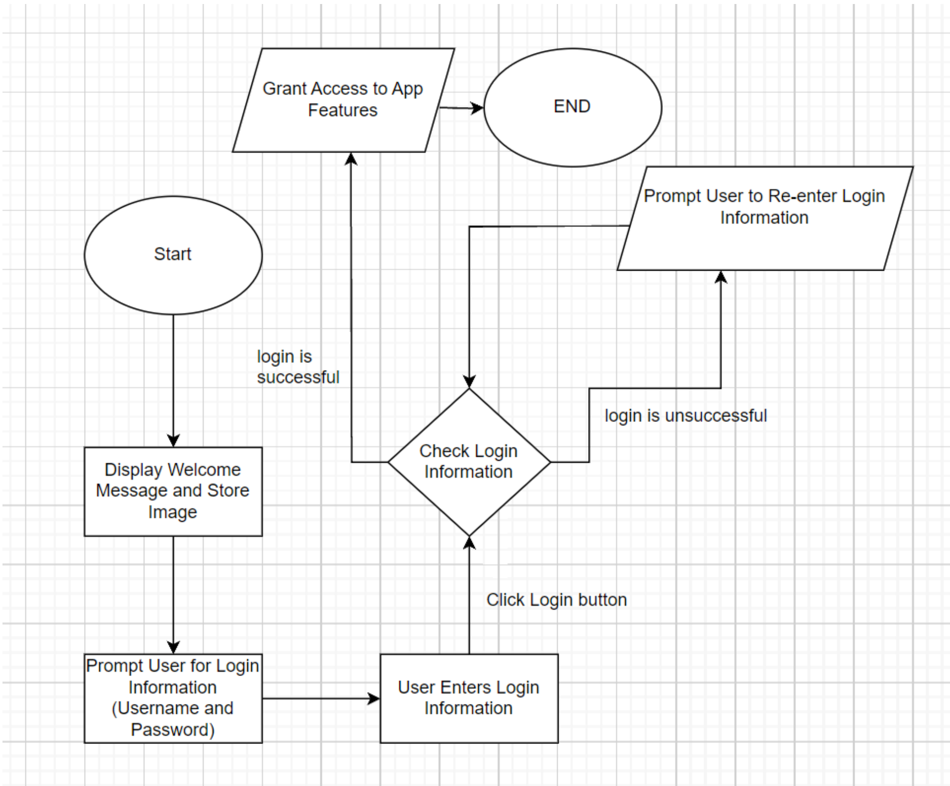
Chapter 5. GUI Creation

In our application, the user interface (UI) serves as the cornerstone of interaction, offering intuitive tools for both customers and employees to navigate and manage various aspects of the system. This section provides an overview of the UI components utilized across different interfaces, including the Manager UI, Employee UI, and Customer UI, each tailored to meet the specific needs of its user base.

5.1. Login, Sign up UI

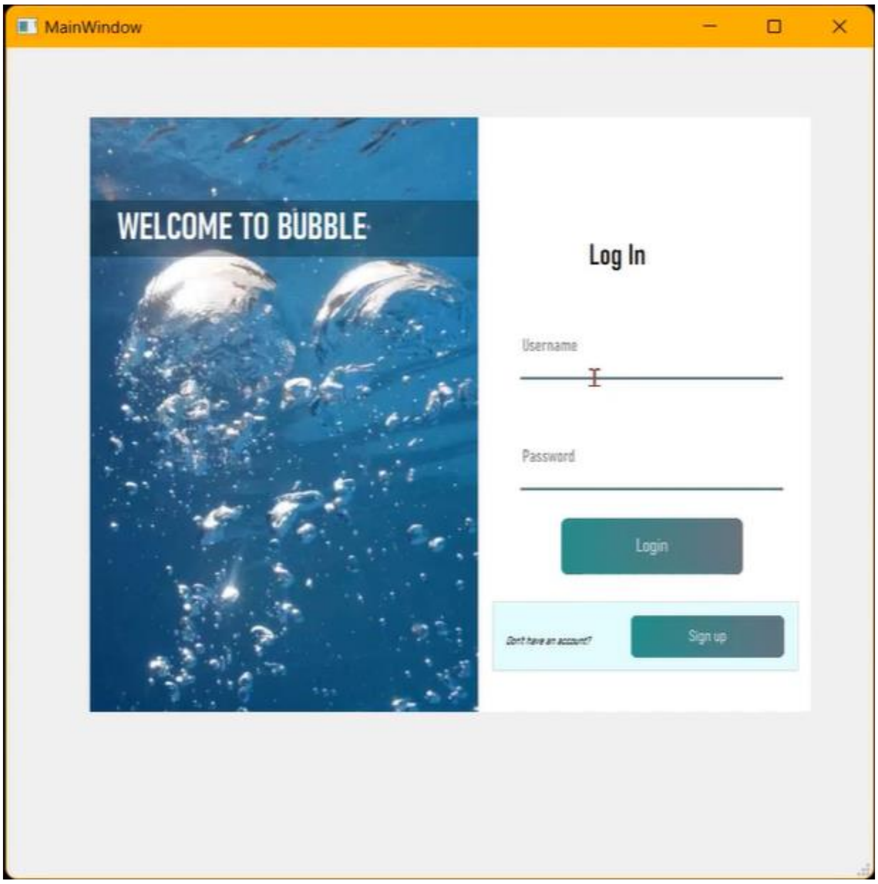
5.1.1. Login

Figure 5.1: Login Flowchart



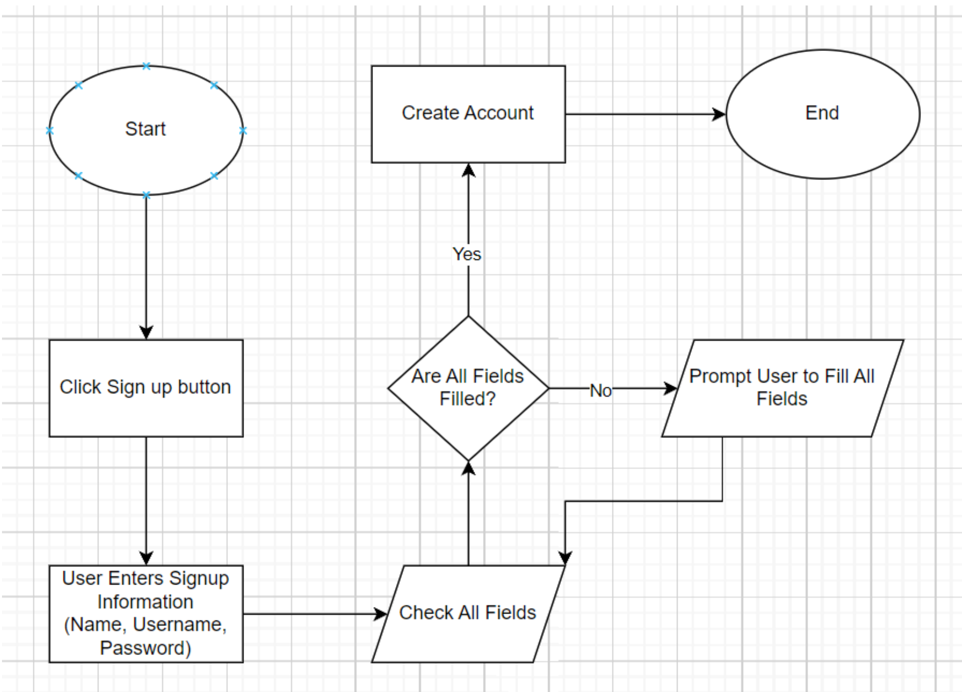
We prioritize user privacy by allowing hidden passwords within the application. On the left side of the user interface (UI), a welcoming message is prominently displayed alongside an image representing the store, facilitating easy identification for customers. Upon successful login, users gain access to the app's features. In case of unsuccessful login attempts, users are prompted to re-enter their login information for authentication. This approach ensures both security and user convenience throughout the login process.

Figure 5.2: Login UI



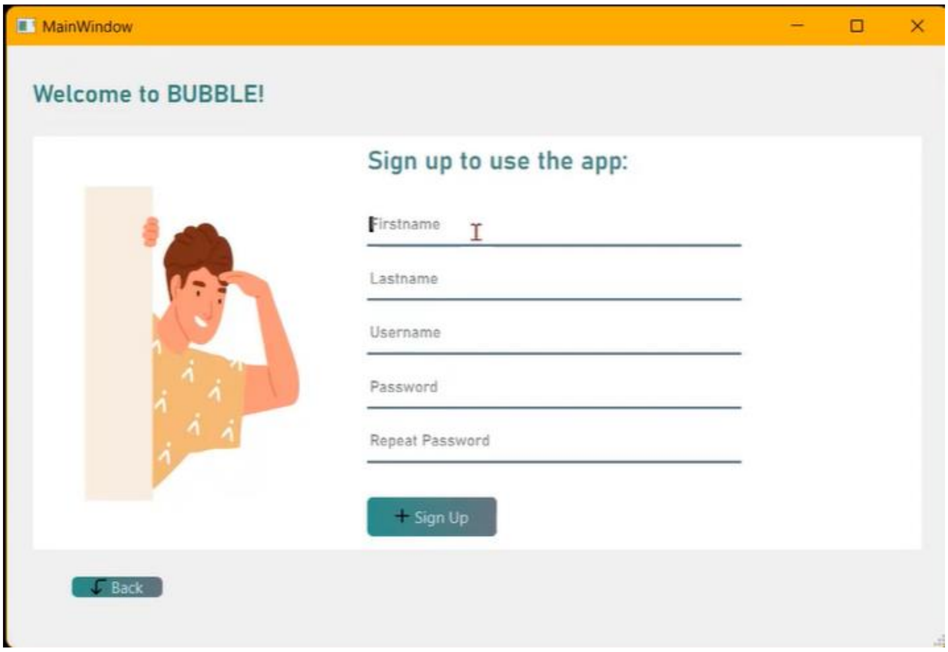
5.1.2. Sign up

Figure 5.3: Sign up Flowchart



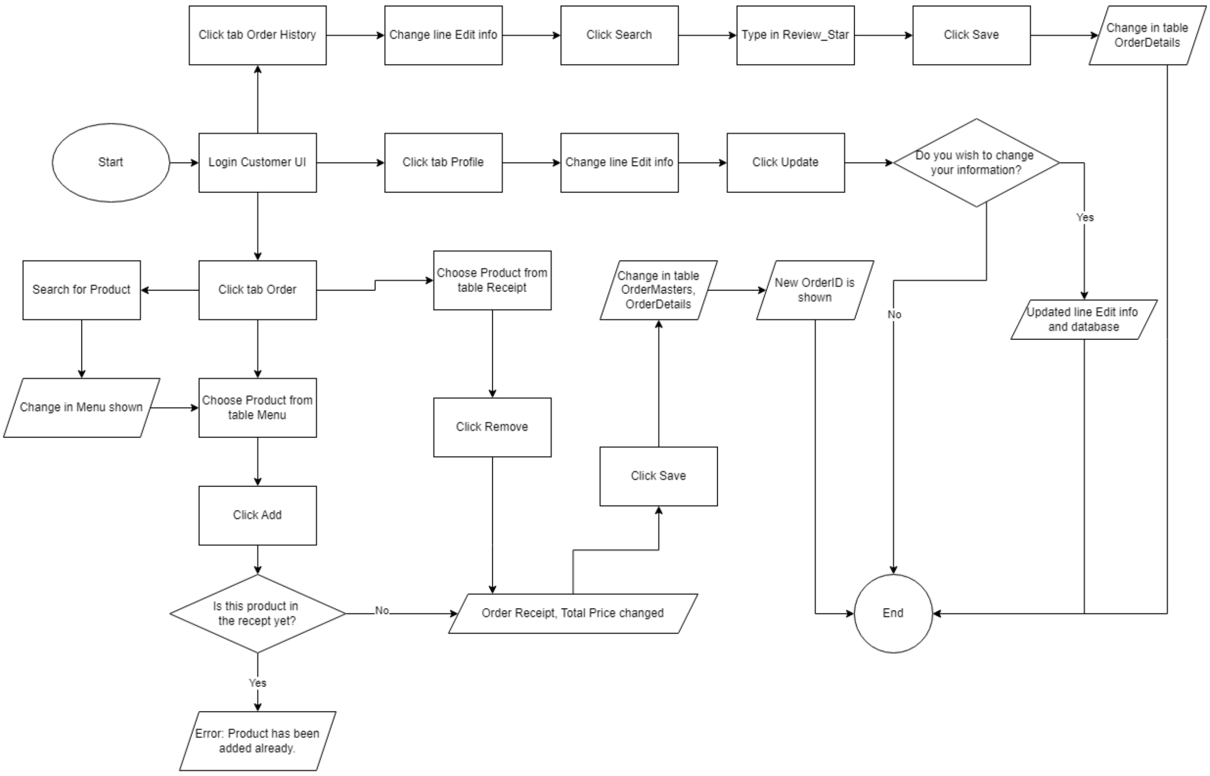
When signing up, users are required to provide their name, username, and password. These fields, represented as line edits, must be completed for the account creation process to proceed. If any of these fields are left unattended, the user will not be able to create a new account. This ensures that all necessary information is provided for a secure and complete registration.

Figure 5.4: Sign up UI



5.2. Customer UI

Figure 5.5: Customer Flowchart



The Customer UI is tailored to meet the needs of users interacting with the application from a customer perspective. This interface offers functionalities for account management, order placement, and order history tracking. Customers can easily update their profile information, browse available products, place orders, and view their order history, all within a seamless and intuitive user experience. By incorporating intuitive design elements and clear navigation paths, the Customer UI aims to enhance user satisfaction and streamline the shopping experience for customers.

5.2.1. Profile

Every customer will occasionally update their account or profile details. To accommodate this, we have included a profile tab where customers can update their information as desired. This tab also displays their unique CustomerID and current Membership status within the store, enabling quicker customer service and access to the latest discounts based on their membership.

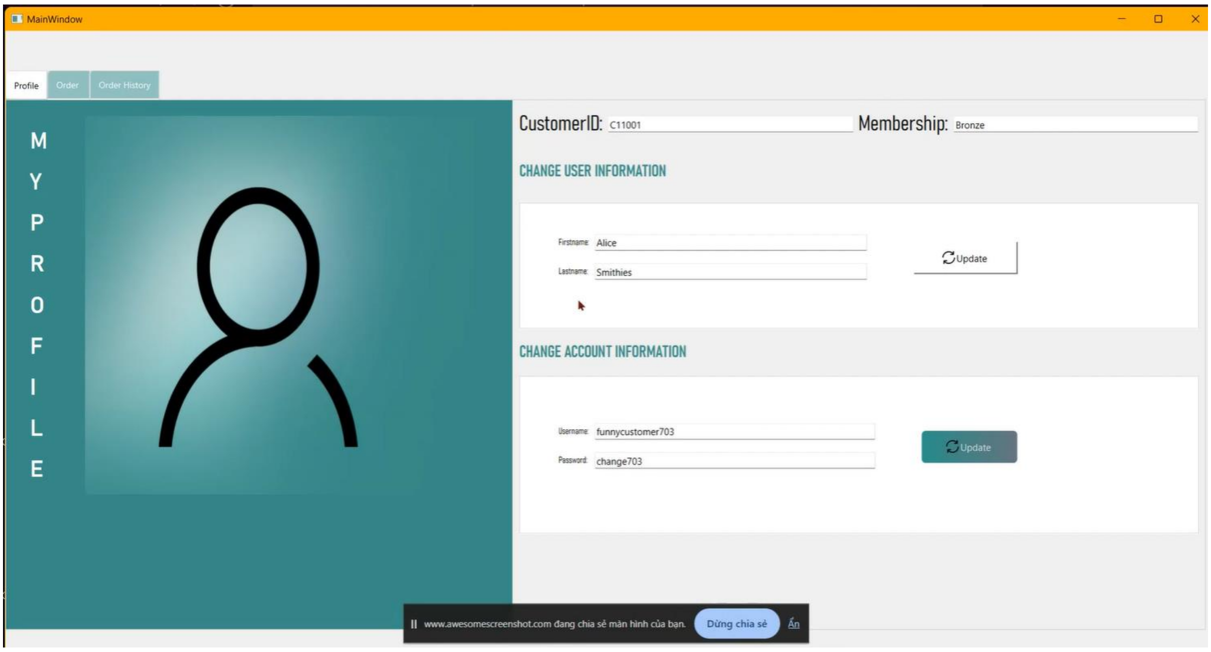
Showing Information

The profile tab prominently displays essential information such as the customer's unique CustomerID and current Membership status. This allows customers to easily reference their account details when interacting with store staff or accessing services, ensuring a more efficient and personalized customer experience. Additionally, this feature helps customers stay informed about their membership benefits and eligibility for exclusive discounts and promotions. The locked Line Edit is used with CustomerID and Customer_Membership, while the rest of the information is left editable.

Updating Information

Using simple functions behind Line Edit widgets, the profile tab also provides a user-friendly interface for updating personal information. This not only helps in maintaining accurate records for the store but also enhances the overall customer experience by ensuring that communications and services are always aligned with the customer's most recent information.

Figure 5.6: Customer UI - Profile



5.2.2. Order

This tab is specially designed for seamless and convenient ordering.

Showing Menu

Through Table widget: Only products available for sales are displayed in this list. Users can also search out products by typing into the search bar.

Through Label widget: Using advanced functions to accompany 19 different product images, we were able to create an interactive flippable menu with Product images.

Placing Order

From the Table widget Menu, the user has to select a row of products they want to add into their new order. Only then will the add function work, else it will announce an error.

After a product has been added into the Order Receipt, the user can change the quantity of that product. Total price is shown on the right corner, automatically changed anytime a product is added or changed in quantity.

Once the customer chooses to save the Order Receipt, both tables OrderMasters and OrderDetails will be updated simultaneously. The newly created order's OrderID is shown on the left of the total price.

Recommendation List

The Table widget Recommender utilizes up to four different recommending functions and will show customers their best fits at any time the app is running. Content-based, collaborative, hybrid, and popular filtering recommendations will automatically alternate based on the characteristics of the customer profile and their present actions.

When users search for products using the search bar, the content-based and hybrid recommendation systems are activated. Clicking the search Button will not only reveal a filtered Menu but also provide new, distinct recommendations based on the search query.

Figure 5.7: Customer UI - Order

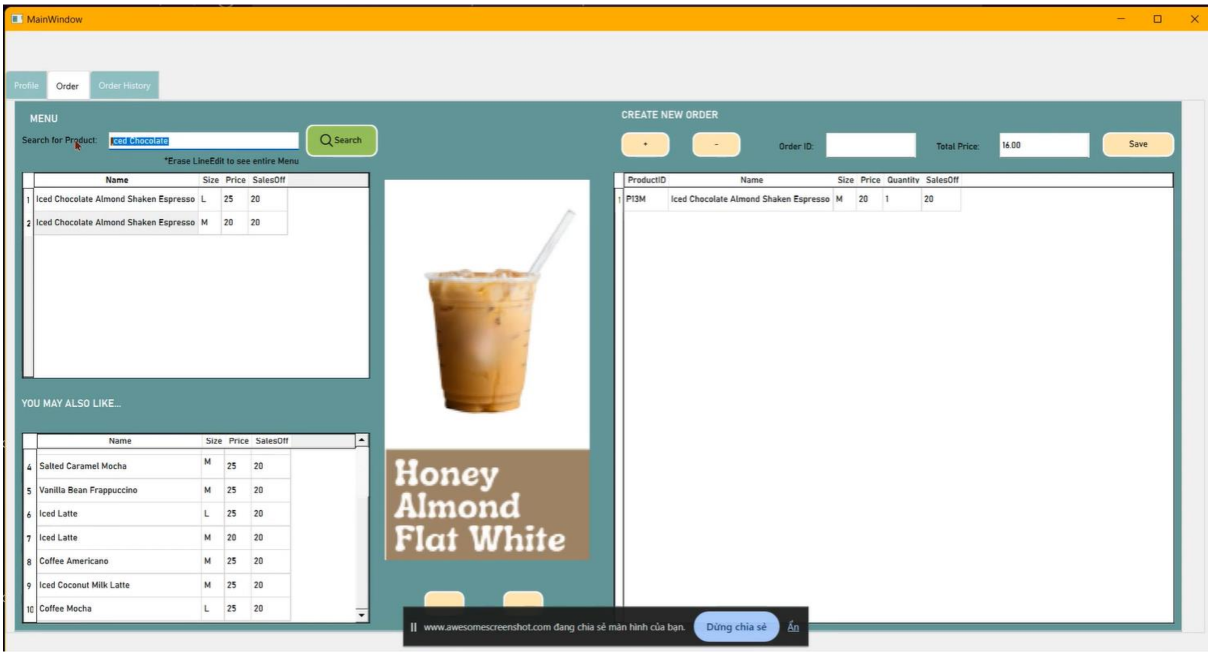
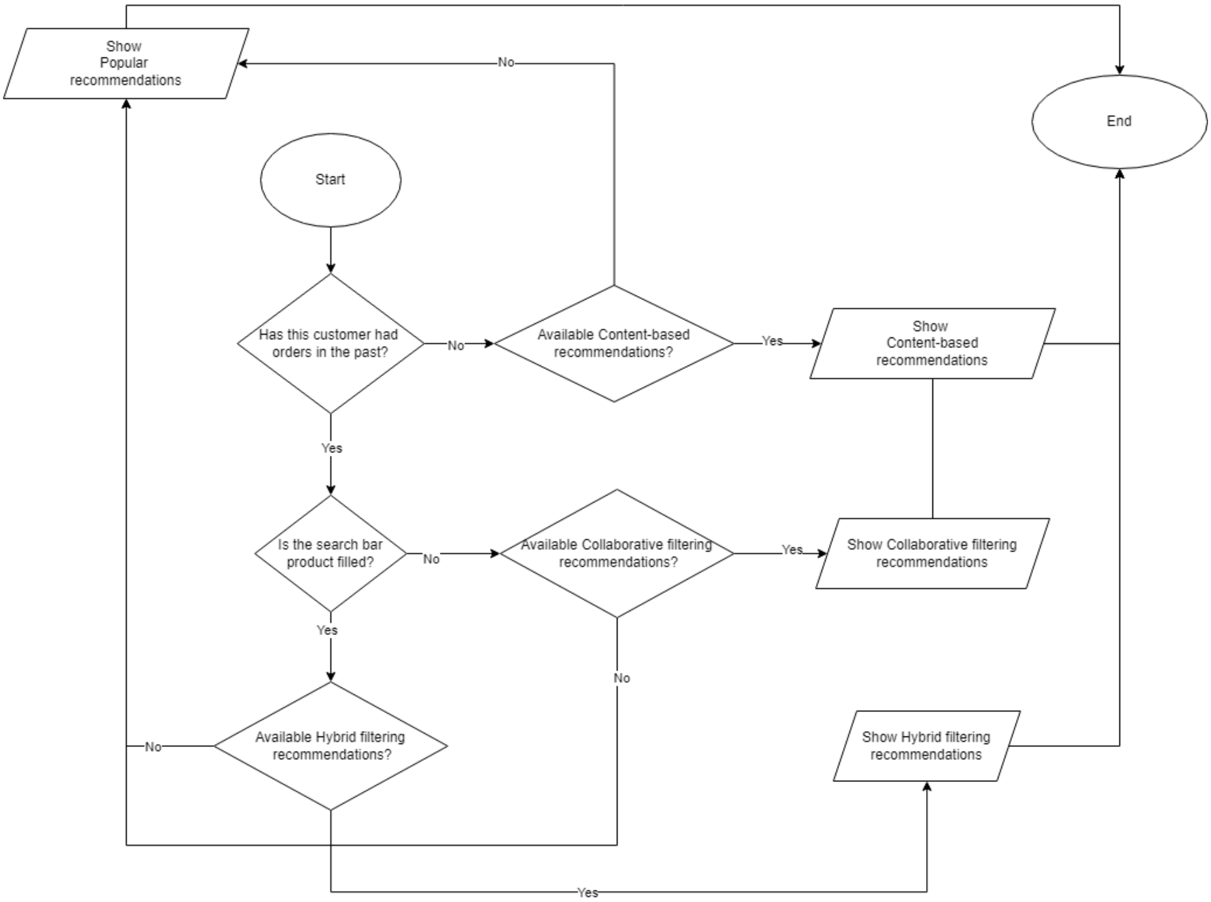


Figure 5.8: The recommender system within Customer UI



5.2.3. Order History

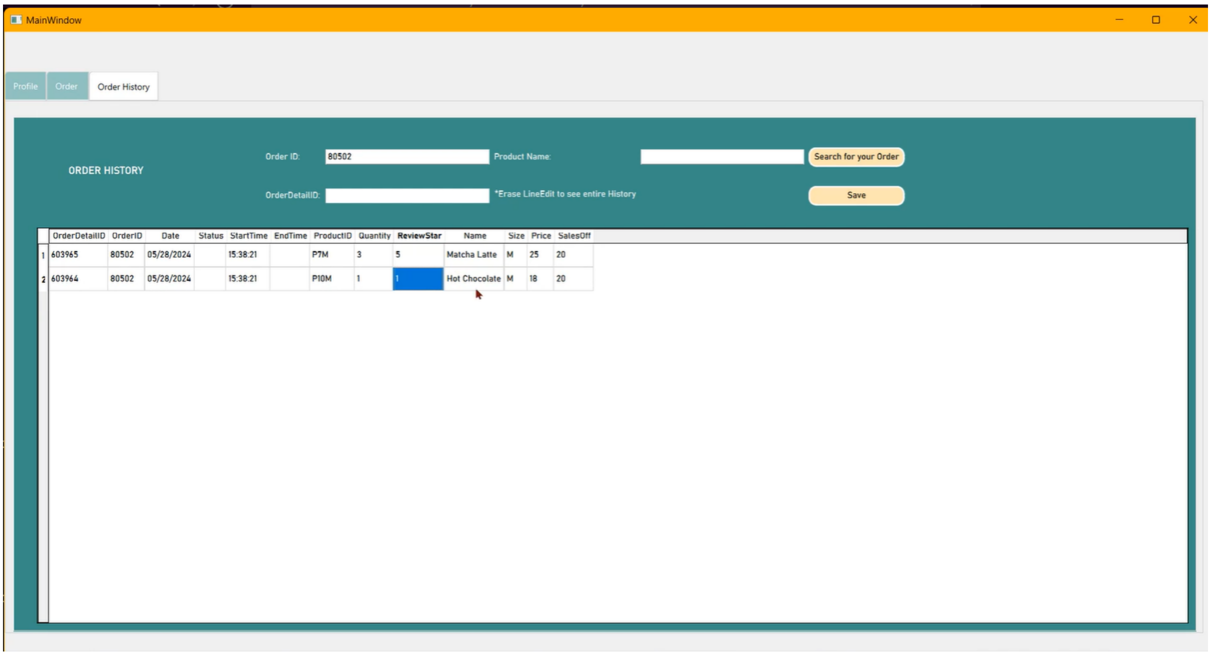
We also included an order history tab for customers to easily search for orders in the past with filtered information. This tab also allows updating Review_Star, making it simple for customers to leave reviews on their experience.

Viewing and Updating Order History

Through table Widget, multiple line Edit for filtered information and search Button, each customer can search for orders of their own in the past.

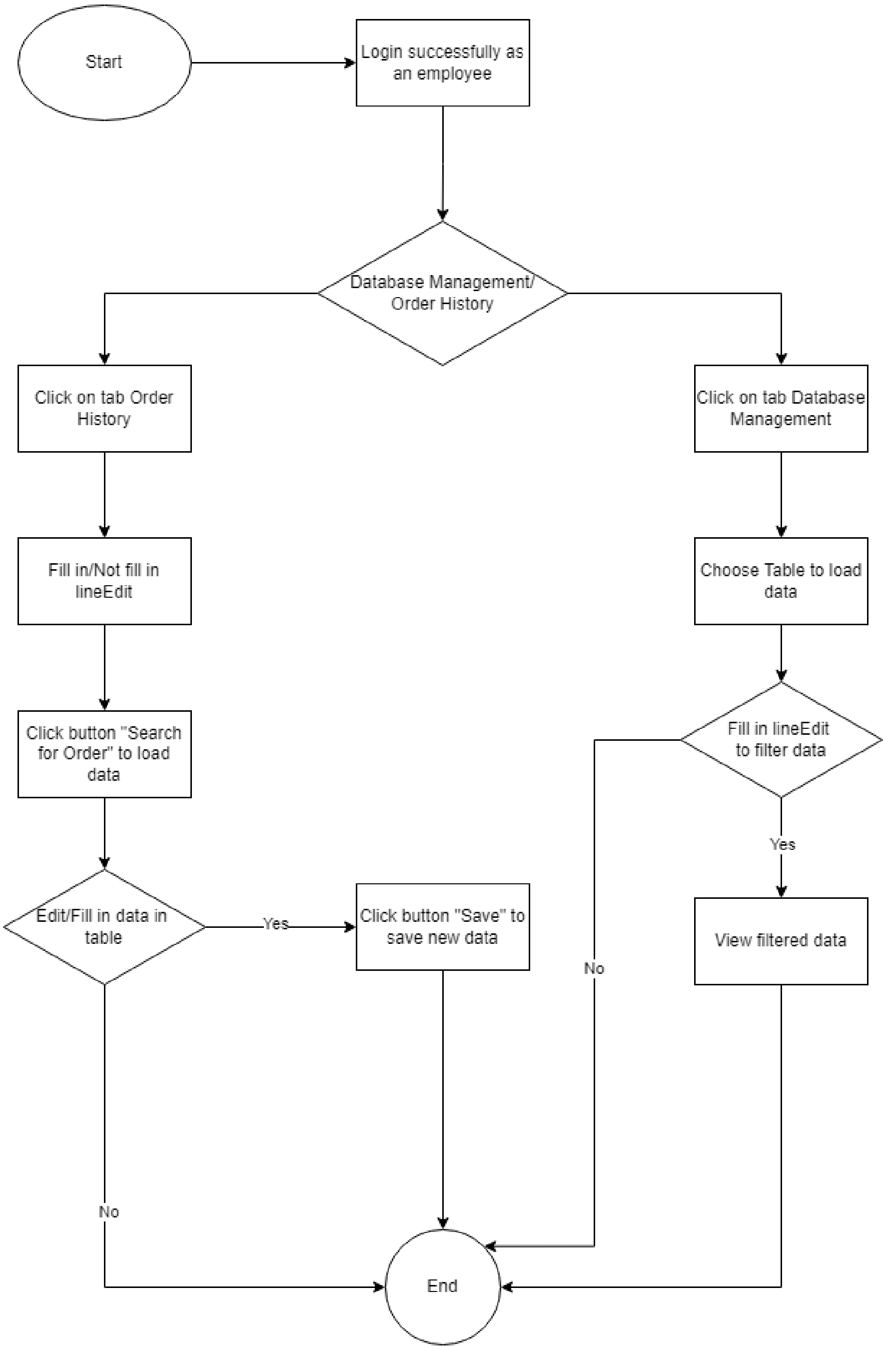
The only column that is editable for customers is Review_Star, allowing them to update their preferences upon each product. After the save review Button is pushed, the Recommendation system will learn this information immediately to put into practice and thereby showing them even better fits in Table widget Recommender.

Figure 5.9: Customer UI - Order History



5.3. Employee UI

Figure 5.10: Employee UI Flowchart



The core functionalities of the Employee UI revolve around efficiently processing orders and providing attentive customer service. This user-friendly interface is designed to manage database operations seamlessly. It offers functionalities to view, search, and update data across multiple tables, including customers, employees, orders, products, and more. Through intuitive design elements such as buttons, table widgets, and line edits, employees can efficiently handle tasks related to order processing, customer management, and inventory control, thereby enhancing operational efficiency and productivity.

5.3.1. Database Management

In this tab, employees can view and search for data about customers, other employees, recipes, products, ingredients, and more. This functionality helps to streamline operations, improve accuracy in order processing, and enhance overall efficiency by providing quick access to essential information.

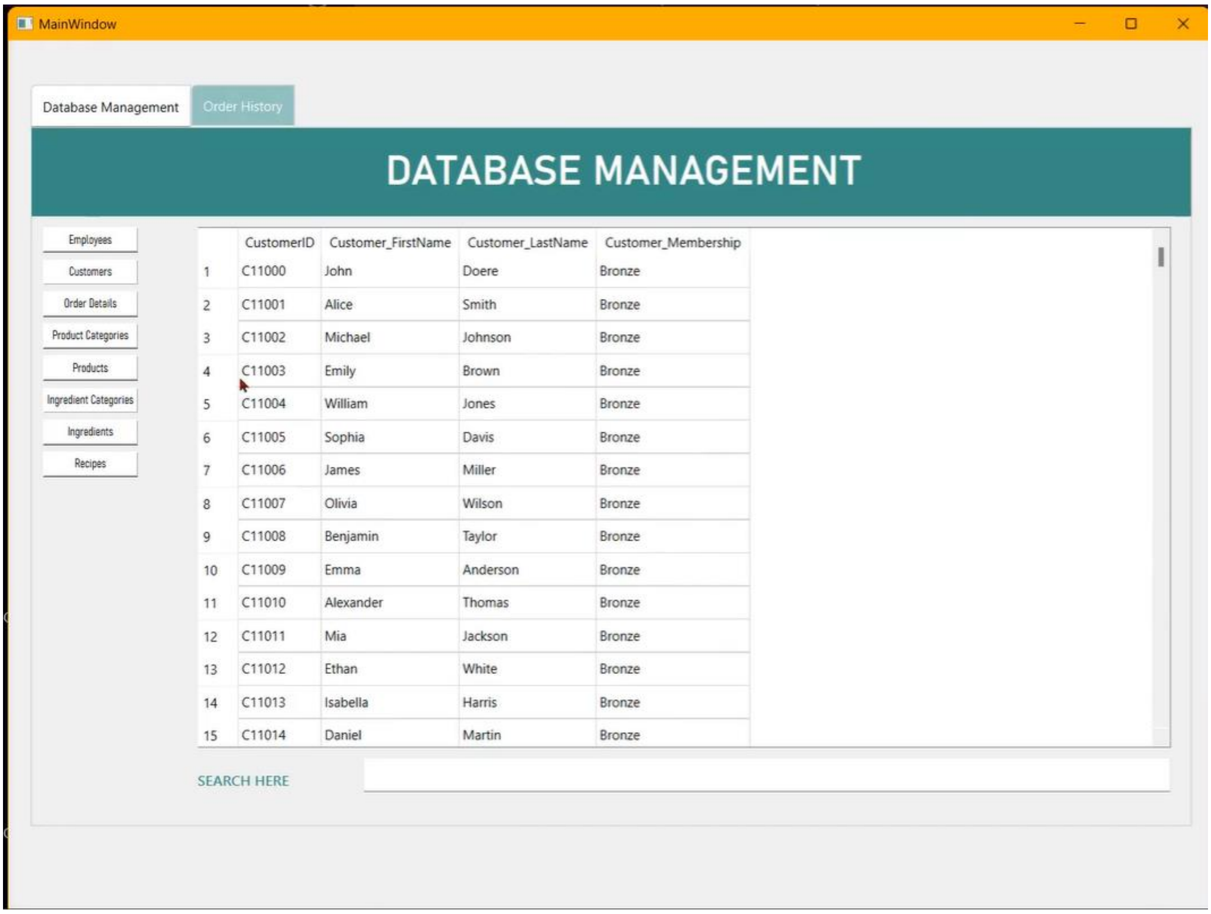
Viewing and Searching Database

Push Button: Employees can click on push buttons with relevant names to open the corresponding table.

Line Edit widget: Employees can fill in this Search area or leave them empty to see filtered or entire tables.

Table Widget: Functions have been added to automatically and smoothly update the view of data in the table according to the change in the line Edit widget.

Figure 5.11: Employee UI - Database Management



5.3.2. Order History

This simple tab is to show OrderMasters and enable staff to update required information each time a new OrderID is created by customers.

Viewing and Searching OrderMasters

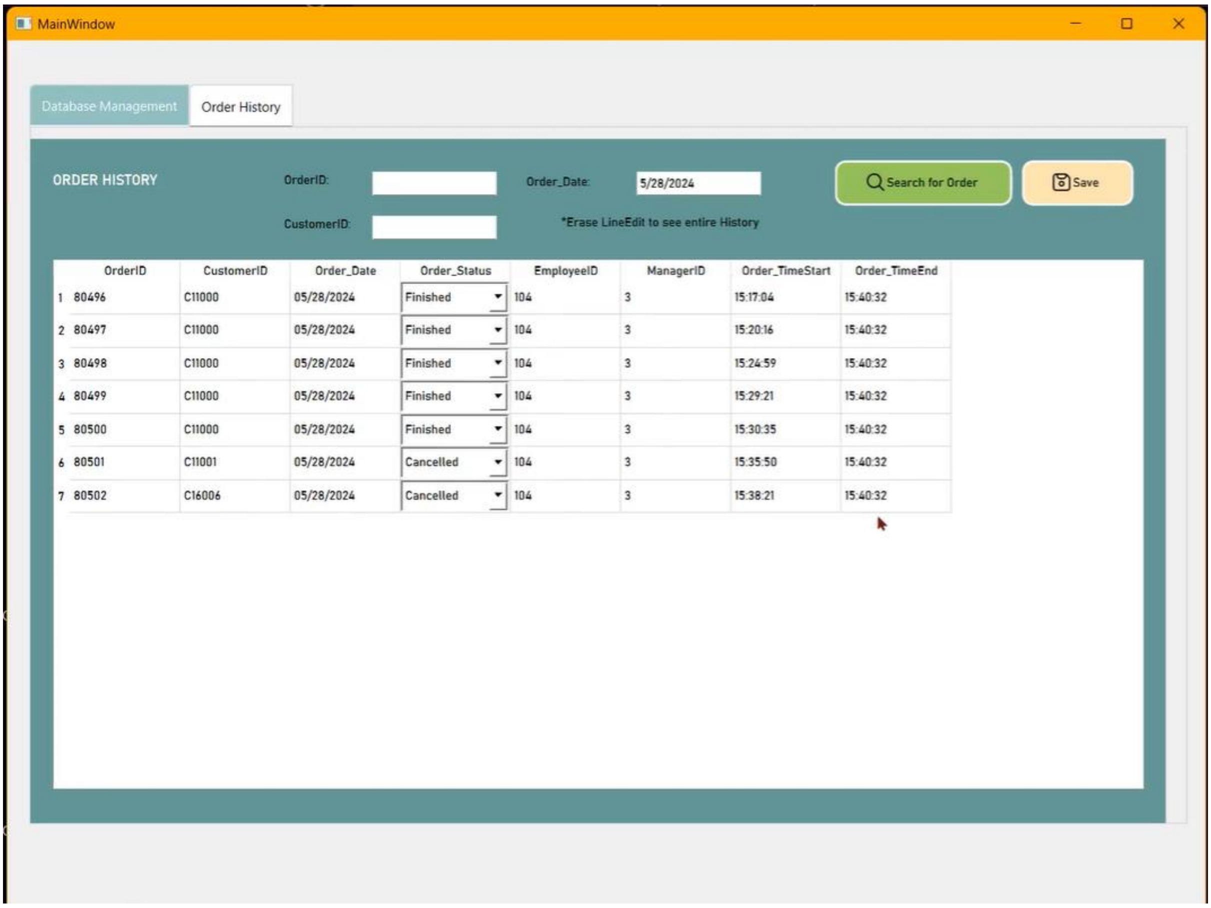
Push Button: Employees can click on push button Search to reload the table to include the newly typed in filtered information.

Line Edit widget: Employees can fill in this Search area or leave them empty to see filtered or entire tables.

Table Widget: Filtered dataframe will be reloaded into this widget each time push button Search is clicked.

Updating OrderMasters

Figure 5.12: Employee UI - Order History



5.4. Manager UI

The Manager UI is meticulously designed to ensure cross-platform compatibility and provide a native user experience. Key components include the tableWidgetStatistic and functional buttons, which enable the generation of insightful charts and visualizations based on sales data. By leveraging libraries such as NumPy and Pandas for numerical computations and data manipulation, and Matplotlib for visualization, the Manager UI empowers users to extract valuable insights and make informed decisions to drive business growth.

5.4.1. Database Management

Managers have full access to editing the entire database apart from accounts and order histories.

Viewing and Searching Database

Push Button: Employees can click on push buttons with relevant names to open the corresponding table.

Line Edit widget: Employees can fill in this Search area or leave them empty to see filtered or entire tables.

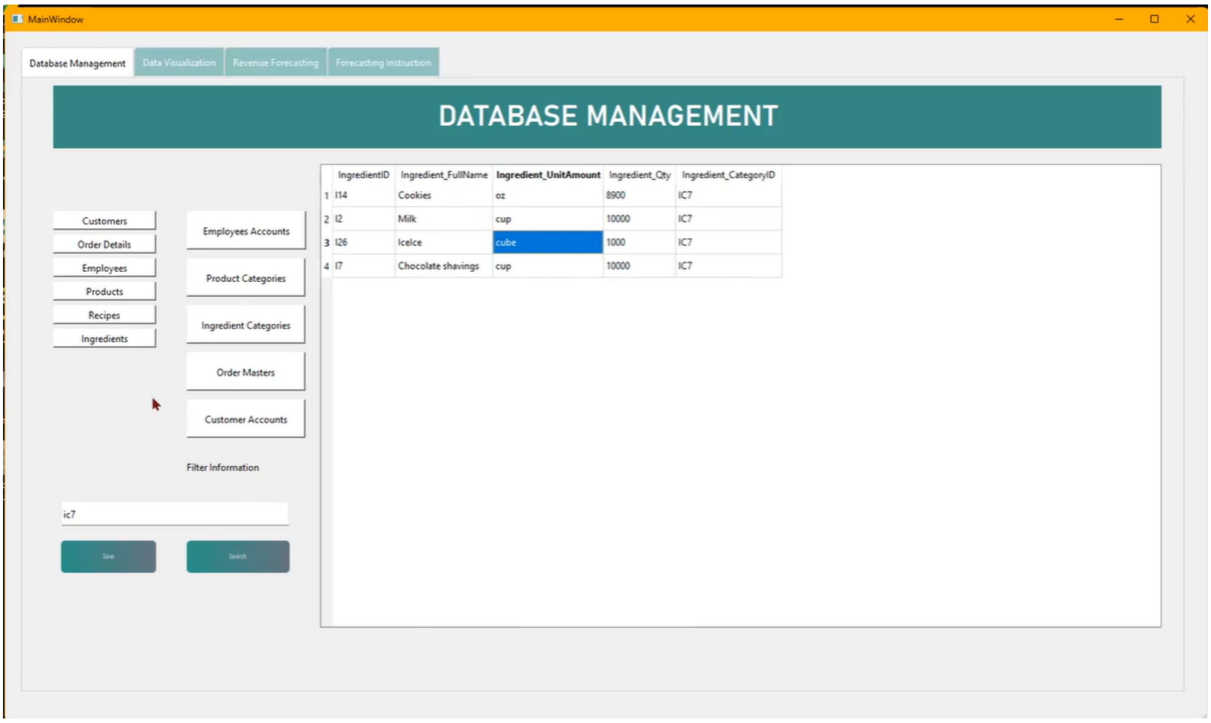
Table Widget: Functions have been added to automatically and smoothly update the view of data in the table according to the change in the line Edit widget.

Updating Database

Table Widget: Managers can update table indirectly through added right click options (add row, remove row) or directly on the cells shown inside Table widget.

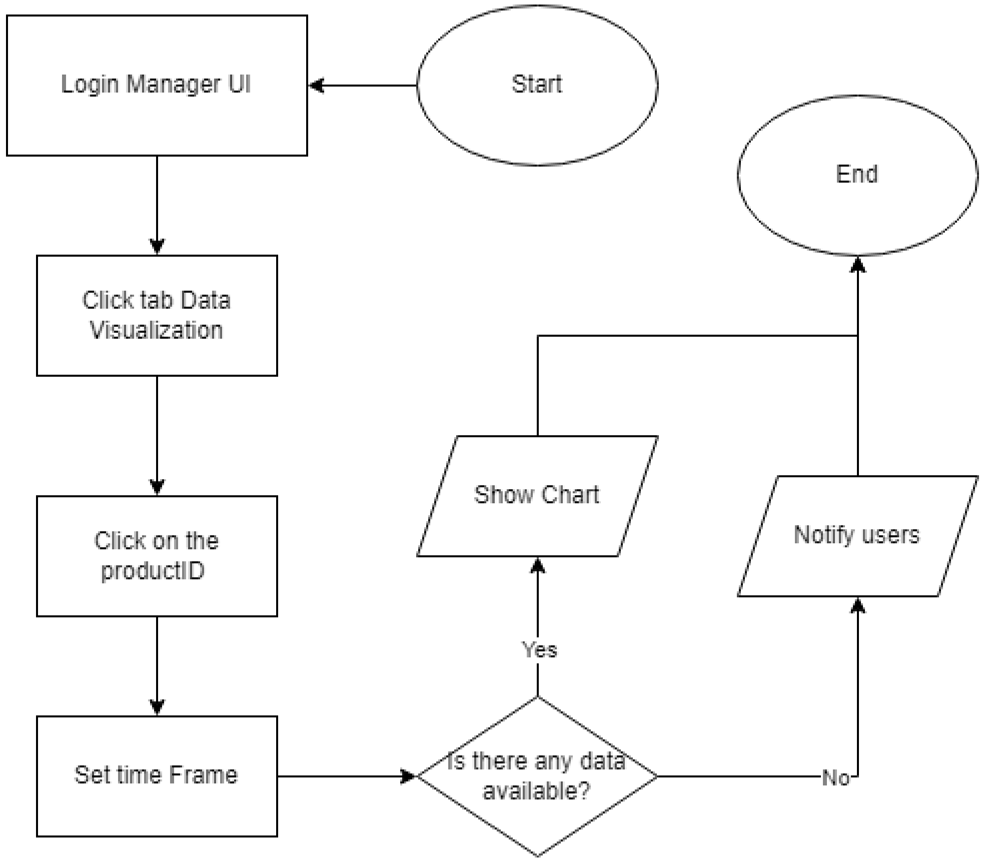
Push Button: In order to actually update the table, managers have to push button Save. If there are errors related to the primary key, the changes will not be updated and the system will notify the manager.

Figure 5.13: Manager UI - Database Management



5.4.2. Data Visualization

Figure 5.14: Manager UI - Data Visualization Flowchart



In order to use data visualizing function, users must first select ProductIDs from the list Widget ProductIDs. This will help filter out unwanted data from other products that managers do not want to involve in the statistics. Having chosen ProductIDs and time frame, the manager can then access 5 distinct charts by clicking each of the 5 different push Button.

Product Count Chart: This is to visualize the number of products sold, showing product popularity.

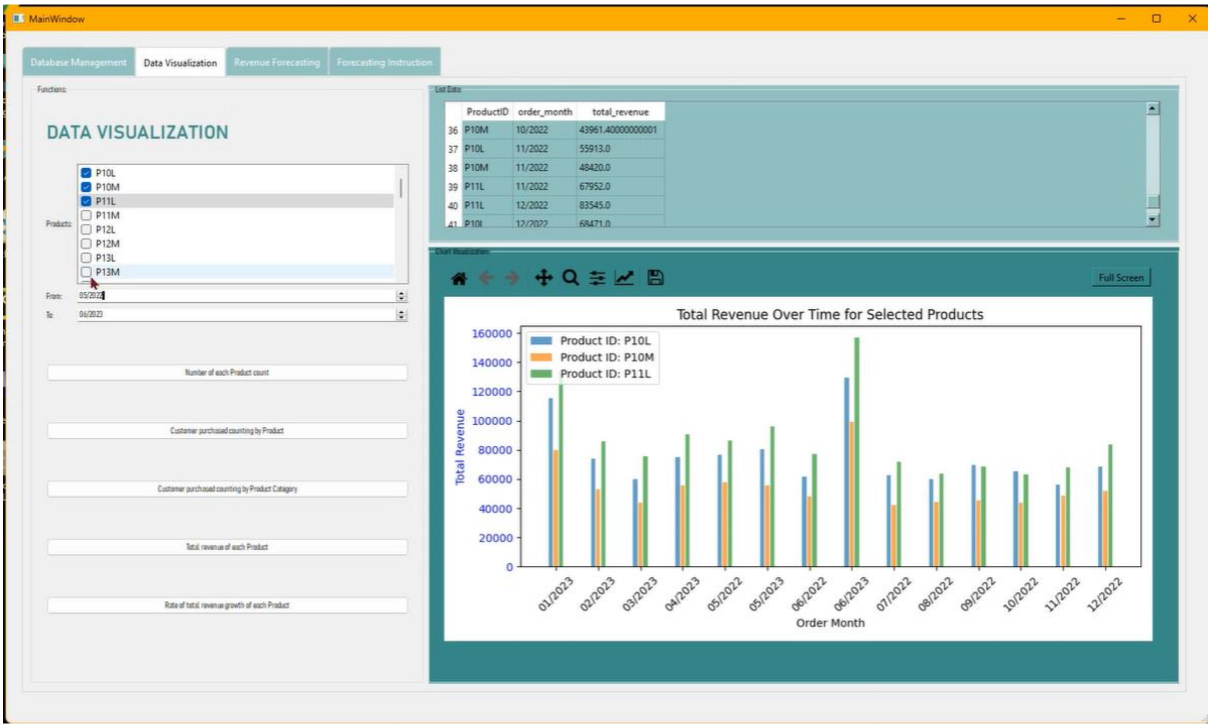
Customer Purchased Count Chart: This reveals how many customers bought each product, highlighting customer preferences and loyalty.

Total Revenue by Product Chart: This illustrates revenue per product, identifying high performers.

Growth Rate by Product Chart: This helps managers examine sales growth over time, tracking performance trends.

Product Category Count Chart: Managers may use this to understand the product distribution across categories, aiding inventory management.

Figure 5.15: Manager UI - Data Visualization

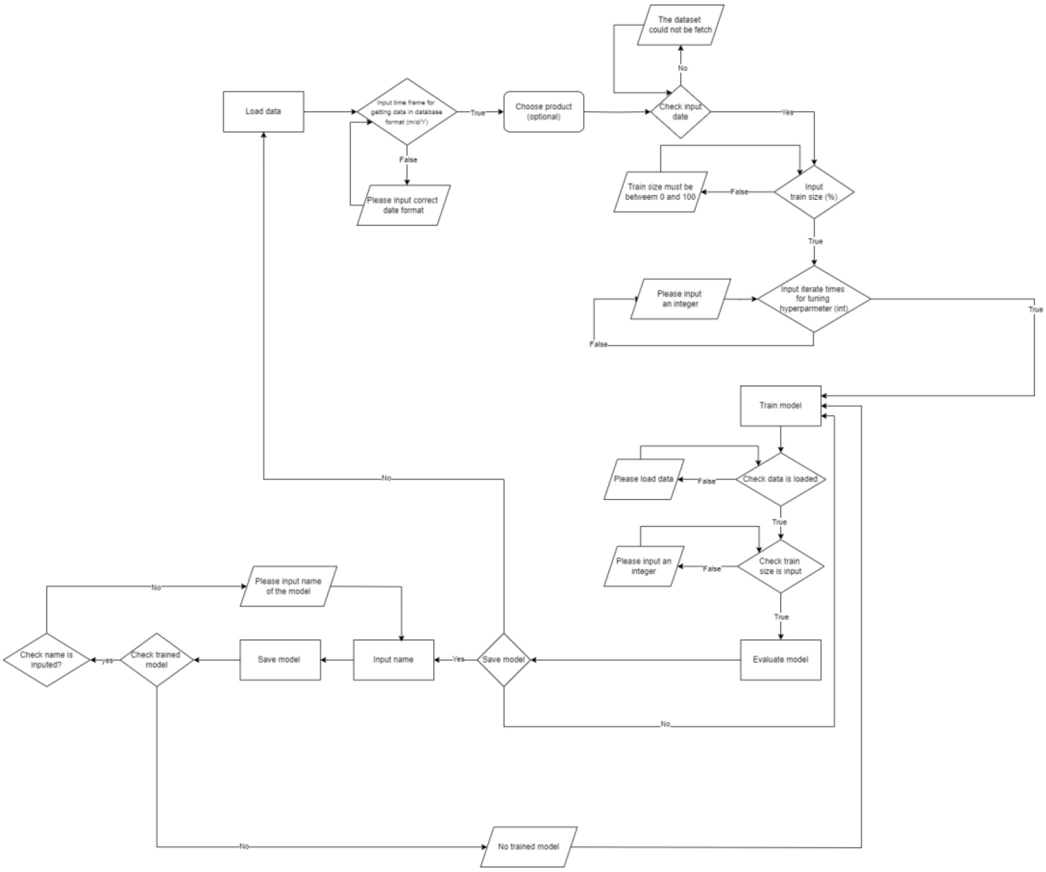


Visual charts provide a comprehensive understanding of product performance, facilitating informed decision-making. They enable the detection of sales growth, customer behavior, and revenue trends, which are essential for strategic planning. Additionally, understanding product category demand helps optimize inventory management. Enhanced marketing strategies can be developed by tailoring campaigns based on customer preferences and top-performing products, leading to better ROI. Continuous business monitoring through tracking key metrics allows for real-time assessment of business health and timely adjustments.

5.4.3. Revenue Forecasting

The Machine Learning tab enables users to forecast future total sales over a specified time frame and monitor sales fluctuations during that period. Users can also train and save custom machine learning models for future use. The user interface includes the following elements:

Figure 5.16: Manager UI - Revenue Forecasting Training, Saving Model Flowchart



Loading Data

Time Frame Selection: Users can specify the start and end dates for the desired time frame using two line edits ‘From’ and ‘To’, the date format should be m/d/Y (e.g., 05/28/2024).

Product Selection: A product combobox on the left side allows users to select the product they want to load data for.

Load Dataset: Clicking the 'Load Dataset' button displays the data in the 'List Data' table widget on the right side. The table includes the following features: ProductID, Product_UnitPrice, Order_Date, and Total_Amount.

Training Model

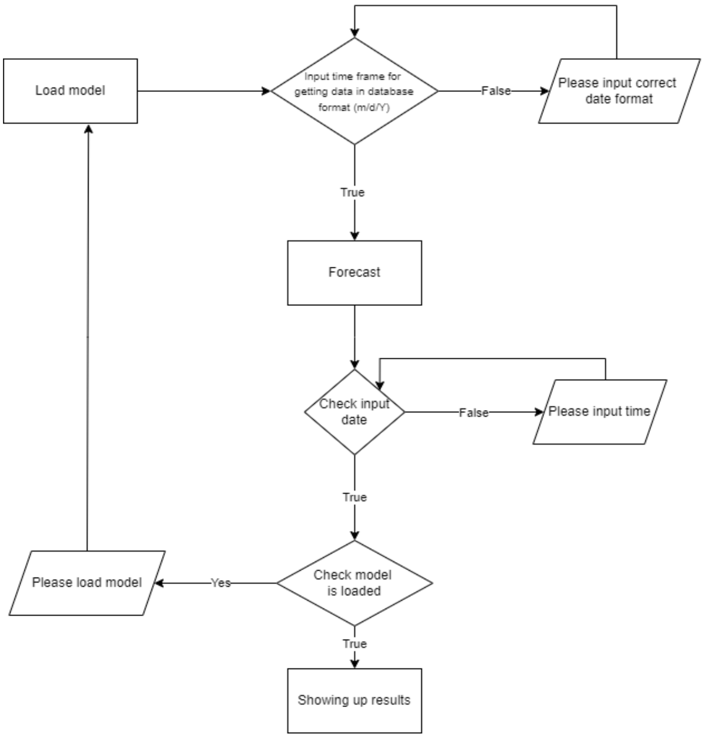
Data Percentage: Users can input the percentage of data to be used for training, ranging from 1% to 100%.

Iteration Count: Users must specify the number of iterations (recommended between 10 and 20) for finding the best model parameters.

Model Evaluation: The evaluation results, including RMSE, MAE, and MAPE, will be displayed in line edits. Lower values indicate better model performance.

Save Model: After obtaining the desired model, users can name the model and save it by clicking the 'Save Model' button.

Figure 5.17: Manager UI - Revenue Forecasting Model Loading Flowchart



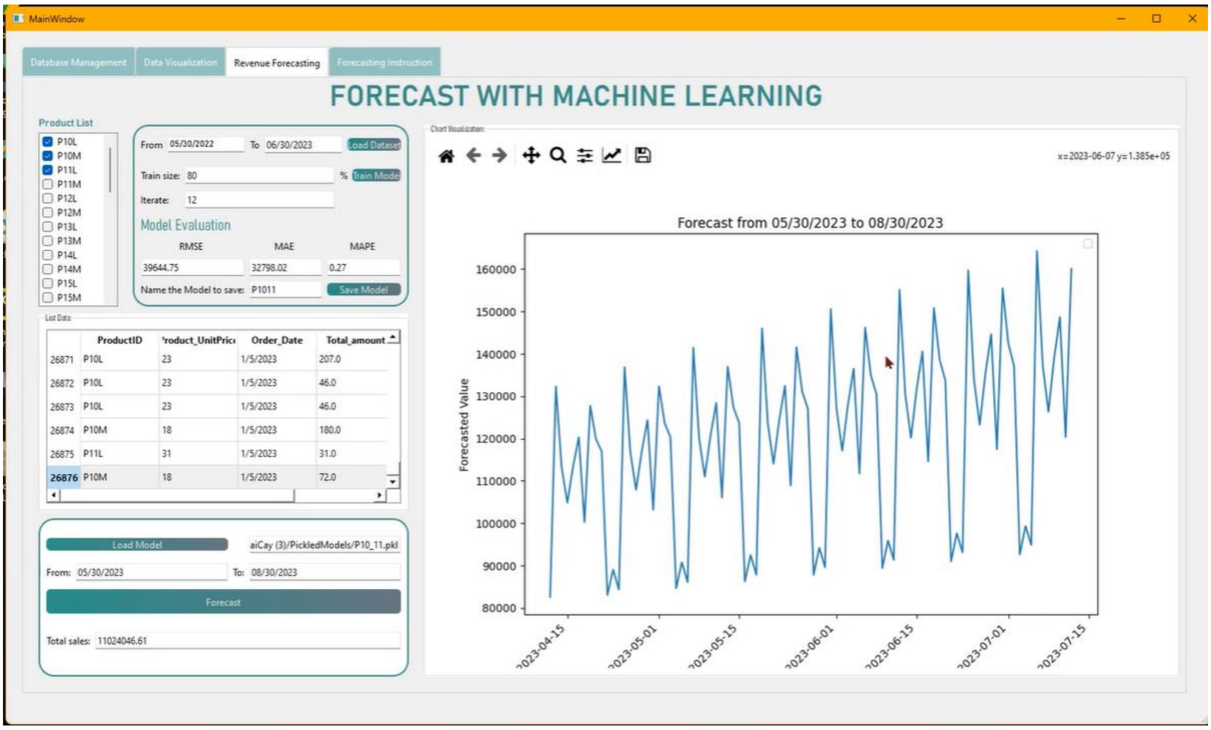
Forecasting

Load Model: Users can load a previously saved model by clicking the 'Load Model' button and selecting the desired model from the saved folder.

Specify Time Frame: Users input the time frame for forecasting in the 'From' and 'To' line edits. the date format should be m/d/Y (e.g., 05/28/2024).

Implement Forecasting: Clicking the 'Forecast' button generates the total sales forecast for the specified time frame. The results are displayed in a chart within the 'Chart Visualization' widget, showing sales fluctuations over the period.

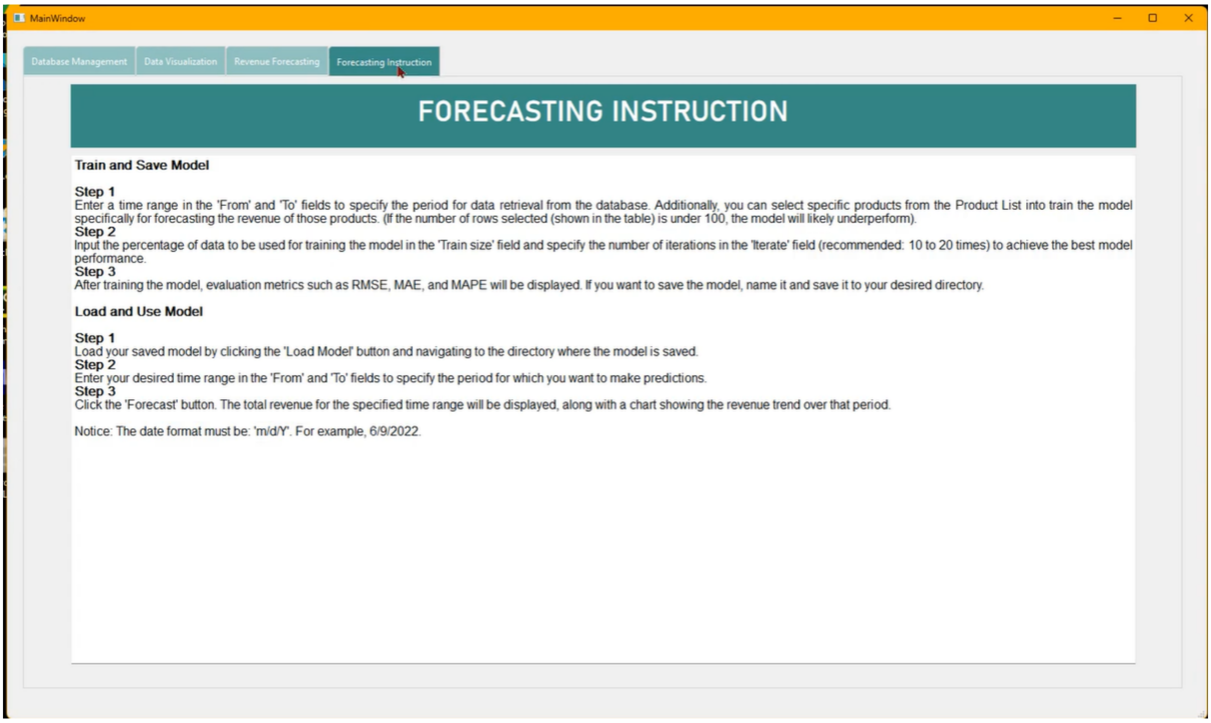
Figure 5.18: Manager UI - Revenue Forecasting



5.4.4. Forecasting Instruction

This simple tab provides essential tutorials to assist managers in operating the app smoothly, helping them to achieve success in business strategizing. Recommended values for certain inputs are advised in prior, ensuring users convenient and swift experience when using the application.

Figure 5.19: Manager UI - Forecasting Instruction



Chapter 6. Conclusion and Future Works

6.1. Conclusion

Nowadays, sales forecasting is an essential part of every industry, particularly for businesses dealing with seasonal items. Accurate forecasting allows businesses to anticipate customer needs and stay ahead in the market. To maintain their market presence, companies must analyze market trends and customer demands early. This enables them to anticipate the goods or services that customers will request and make the necessary preparations to supply them in advance. In this study, we present a sales forecasting model designed to predict future product demand based on sales revenue. Additionally, we propose a collaborative recommendation system to predict user preferences and provide personalized recommendations.

In the forecasting segment, we conducted an experimental analysis of five models: ARIMA, SARIMA, Exponential Smoothing, Random Forest, and XGBoost, applied to forecast daily sales for a small business. Our findings indicate that most of these models outperform classical forecasting methods. Notably, the SARIMA model stood out as the best-performing model, achieving an RMSE of 9898.76, an MAE of 8300.07, and a MAPE of 0.089. It produced quick and accurate forecasts, demonstrating superior performance compared to the other four models.

For the recommendation system, we tested three models to identify the most suitable recommendation algorithm. Our experiments revealed that content-based filtering and hybrid methods effectively generate recommended product lists by combining a person's purchase history with new selections. Meanwhile, the collaborative filtering model excels in introducing products to individuals with no purchase history. The main advantage of these models is their ability to provide immediate and continuous recommendations for all consumers, whether they have a specific product in mind or no order history at all.

6.2. Future Works

The proposed forecasting model and recommendation system have demonstrated promising results in predicting sales and recommending relevant products to customers. However, model performance depends significantly on the quality and size of the training dataset. A model that performs well on one dataset may not do so on another. To address this, future improvements could include expanding the range of input parameters, such as economic indicators, social media trends, and competitor activities. Additionally, incorporating a wider variety of machine learning algorithms would allow users to select the most suitable model for their data.

Acknowledging the importance of customer segmentation, we have integrated essential data into the store's database for this purpose. This data encompasses several tables, notably

OrderMasters, OrderDetails, and Products. Initial steps involve the removal of rows with a status of "Cancelled" in the Order_Status field, indicating incomplete transactions. Subsequently, key customer metrics are extracted, including their unique identifier (CustomerID from OrderMasters), most recent purchase dates (maximum Order_Date for each unique CustomerID from OrderMasters), purchase frequency (count of unique CustomerID in OrderMasters), and total expenditure (computed by retrieving Product_UnitPrice from the Menu table based on ProductID and Product_Qty from OrderDetails, and adjusting for any PriceChange). Upon completion of data acquisition, preprocessing procedures are implemented to refine the dataset for subsequent customer segmentation analysis. An understanding of customer behavior derived from this process can significantly enhance targeted marketing strategies, thereby fostering customer satisfaction and retention.

Enhancing data visualization capabilities would make the system more user-friendly and informative. Adding more chart types, such as interactive scatter plots and heat maps, can provide a more comprehensive understanding of the data. Integrating sales data with web traffic data from Google Analytics would offer deeper insights into customer behavior.

Finally, the employee-facing GUI could be improved with advanced filtering and search capabilities, enabling employees to quickly find specific information. These enhancements will lead to better decision-making and increased sales performance by providing users with deeper insights and more accurate forecasts.

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MATERIALS